

THE METHOD 1.6 — SPECTRA COMPENDIUM

The index of Rydberg channels, and a value for every cell of it.

This compendium supplies the book with coordinates. Its subject is Λ_{spectra} ; its output is a defect for every channel of every element, graded by how the value was got. *The register carries the working record — the corrections, the reversals, and what remains open. This carries the answers.*

0 • THE COORDINATE SUPPLY

What a coordinate is

$(Z, \text{charge}, \ell, 2S+1) \rightarrow \delta$, with a grade, a source and a bound

Z the atomic number • **charge** the ionisation stage, 1 for the neutral • **ℓ** the Rydberg orbital • **$2S+1$** the multiplicity, which Hund's rule on the core derives from the ground state and does not require measuring.

δ is the quantum defect: $E = -Z_c^2 R / (n - \delta)^2$, so a channel's whole Rydberg series follows from one number and the core charge.

DECLARATION — the stored δ is the MEDIAN over a cell's members (T2, closed). This was measured, not chosen: of 433 series whose member count reproduces exactly from raw, the stored value equals the median in 430 (99.3%); the 18 mean-agreements are series where median and mean coincide and distinguish nothing (R 1673, 1675, enlarged sample R 1679). The mean and asymptotic perspectives are carried alongside per cell, as perspectives of one definition (T6, M's ruling, session 1.7.3).

The grades

grade	rows	what it means
exact	929	forced by symmetry — $\delta = 0$ when one electron faces a bare nucleus
measured	358	fitted to captured levels against an ionisation limit
computed	103,545	supplied by the channel equation, within the domain stated below

Nothing is ungraded. A value with no provenance is not a coordinate.

The witness

*Register 1578. This table used to grade cells VERIFIED, POSSIBLE and IMPROBABLE. **Improbable is a judgement about the future, not a fact about the record** — and register 1287 had already established the rule it breaks: an index whose coordinates mix the OBJECT with the OBSERVER cannot close. What replaces it is a fact and a named obstacle.*

cells		
the index admits	104,832	every channel of every element
witnessed	358	reality has been consulted for this cell
unwitnessed	104,474	it has not — see the bound

0.341% of the index has been checked against reality. Every other cell is theory, and the compendium should be read that way.

The bounds on the unwitnessed

Not probabilities. Named obstacles, each of which may or may not be lifted.

cells	the obstacle
28,526	series unresolved above ng in any published analysis
24,312	no long-lived isotope
17,626	not keyable: no single $2S+1$ (hole+electron or multi-valence)
11,605	open-shell core, 16 parents

cells	the obstacle
9,756	open-shell core, 3 parents
5,280	open-shell core, 119 parents
2,691	no analysis located at this charge (NOT a bound on existence)
1,760	no nuclide synthesised; theoretically admitted — Janet left-step, 8s(2) closes the eighth period at element 120
1,744	no nuclide synthesised; theoretically admitted — Janet left-step, 8s(1) opens element 119
929	derived by symmetry; no measurement required
240	none — separable series, simply not yet measured
6	limit 41449.451; one electron outside a closed shell
5	no primordial isotope (R 1587)
5	limit 31406.4677325; one electron outside a closed shell
3	limit 66928.04; one electron outside a closed shell
3	limit 49266.66; one electron outside a closed shell
2	limit 48278.48; one electron outside a closed shell
2	limit 48387.634; one electron outside a closed shell
2	limit 32848.872; one electron outside a closed shell
1	limit 46670.107; one electron outside a closed shell
1	limit 43762.6; one electron outside a closed shell

The charge bound is the one to distrust. It reads *no analysis located**, *not* no analysis exists* — and register 1577 found twelve MEASURED cells above it in this index's own file (Ti XI, Fe XV, Fe XVI), with Theodosiou 1986 covering all positive ions of all atoms to $Z = 37$.

Accuracy figures in this compendium are quoted against the WITNESSED-plus-possible set, not against the whole index.

The bound, which needs no measurement at all

$$B = \min(p, n_0 - \ell - 1)$$

with p the core's orbital count at that ℓ and n_0 the first Pauli-allowed principal number, both read from the ground-state configuration. **$\text{floor}(\delta) \leq B$ holds on every measured channel without exception**, and B travels with every row of the table — so the book can bound a defect without consulting one.

The filler, and its domain

Where no capture reaches, the channel equation supplies the value:

$$p > 0 : \delta = 0.3772 \cdot p^{\frac{1}{2}} e(N_e) \cdot N_e^k \cdot \ln(c+1)/c \quad p = 0 : \delta = 0.5415 \cdot C(Z) \cdot (N_e - 1)/N_e \cdot N_e^k \cdot \ln(c+1)/c$$

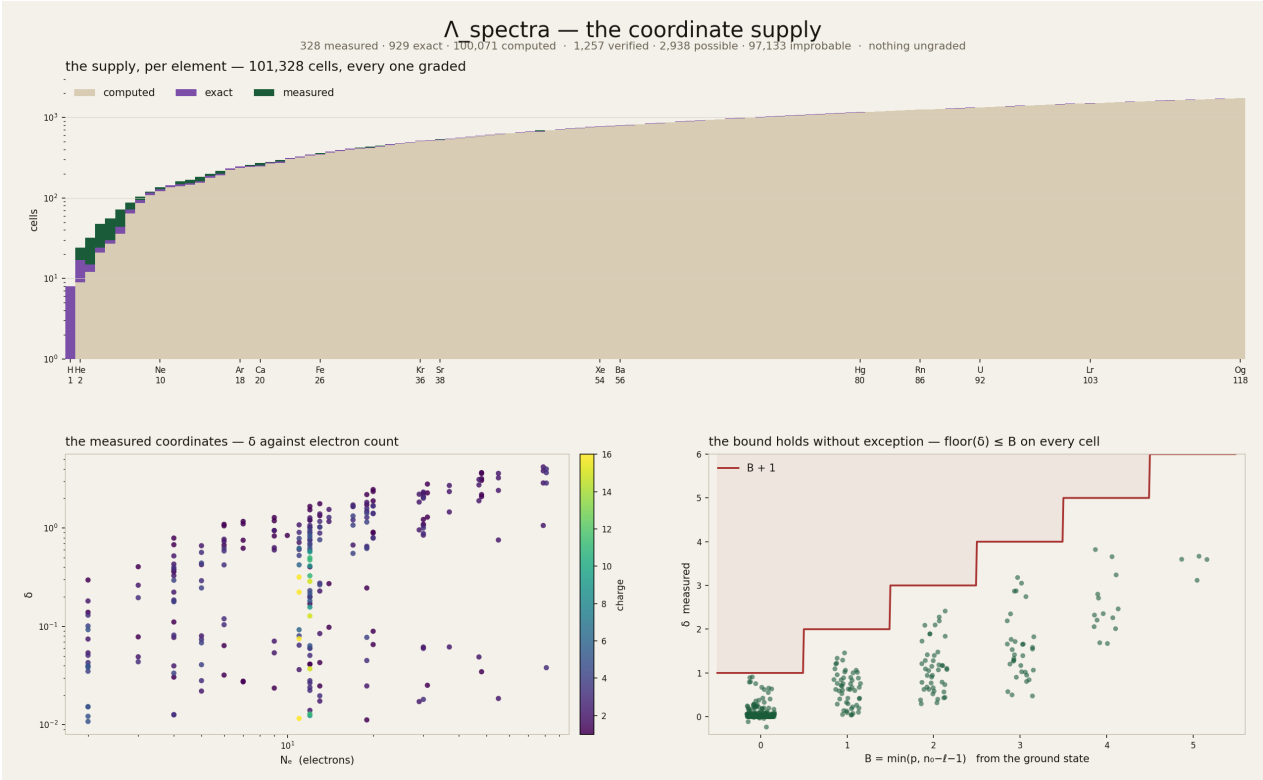
$$e(N_e) = 0.8297 - 0.0900 \cdot \ln N_e \quad k = 0.4942 \quad N_e = Z - \text{charge} + 1 \quad C(Z) = \text{the collapse coordinate across the Janet block boundary}$$

Domain: $Z \leq 92$, charge ≤ 10 , $\ell \leq 4$, a single-parent core. Fitted there on 277 channels at rms 0.1329 and R^2 0.9747, anchored at $Z = 2$ and $Z = 90$. *Every value outside the domain is marked improbable in the status column.*

The table

COORDINATES.tsv — 104,832 rows:

Z charge l mult delta grade source B status



The coordinate supply

Above: the supply per element, on a log scale — the measured and exact cells against the computed. Below left: every measured coordinate, δ against electron count, coloured by charge. Below right: the Pauli bound, which holds on every measured cell and is read from the ground state alone.

I • THE INDEX

The four coordinates, and where each comes from

coordinate	alphabet	derived from
Z	1 ... 118	given
charge	1 ... Z	given
ℓ	0 ... 7	given
2S+1	Hund's rule on the core	the ground state

The multiplicity is not a free coordinate. Hund's first rule applied to the core's ground configuration gives the allowed values, exact on all 24 electron counts tested. Applying it as a constraint removes **247,248 cells** — **71%** of what the naive product would admit.

Which cells exist

A cell is admitted when charge $\leq Z$, at least one electron remains, the multiplicity is one Hund's rule allows, and an orbital of that ℓ is available. Every cell the index admits is physically possible: the forbidden ones were excluded at construction, so the survey holds none.

Which cells can be measured

restriction	cells
the index admits	104,832
$Z \leq 92$ — naturally occurring	61,152

restriction	cells
and charge ≤ 10	11,416
and $\ell \leq 4$ — where series resolve	4,395
and a single-parent core	1,755

The parent-term wall

A closed-shell core has one parent term and gives one Rydberg series per ℓ . An open-shell core gives one series per parent, all interleaved, converging on different limits.

Fe IV's $3d^4$ core carries sixteen LS terms. Its published levels show thirteen of them across **24 distinct (parent, ℓ , term) series, every one with a single member**. Fe IV has about a thousand analysed levels and no extractable defect: *the levels are identified and the series are not separable*.

This is why the compendium holds Ne-like and Na-like ions at charge 15 and 16 and no open-shell ion above charge 6.

An open-shell core does not produce the object a quantum-defect index holds.

The Janet collapse

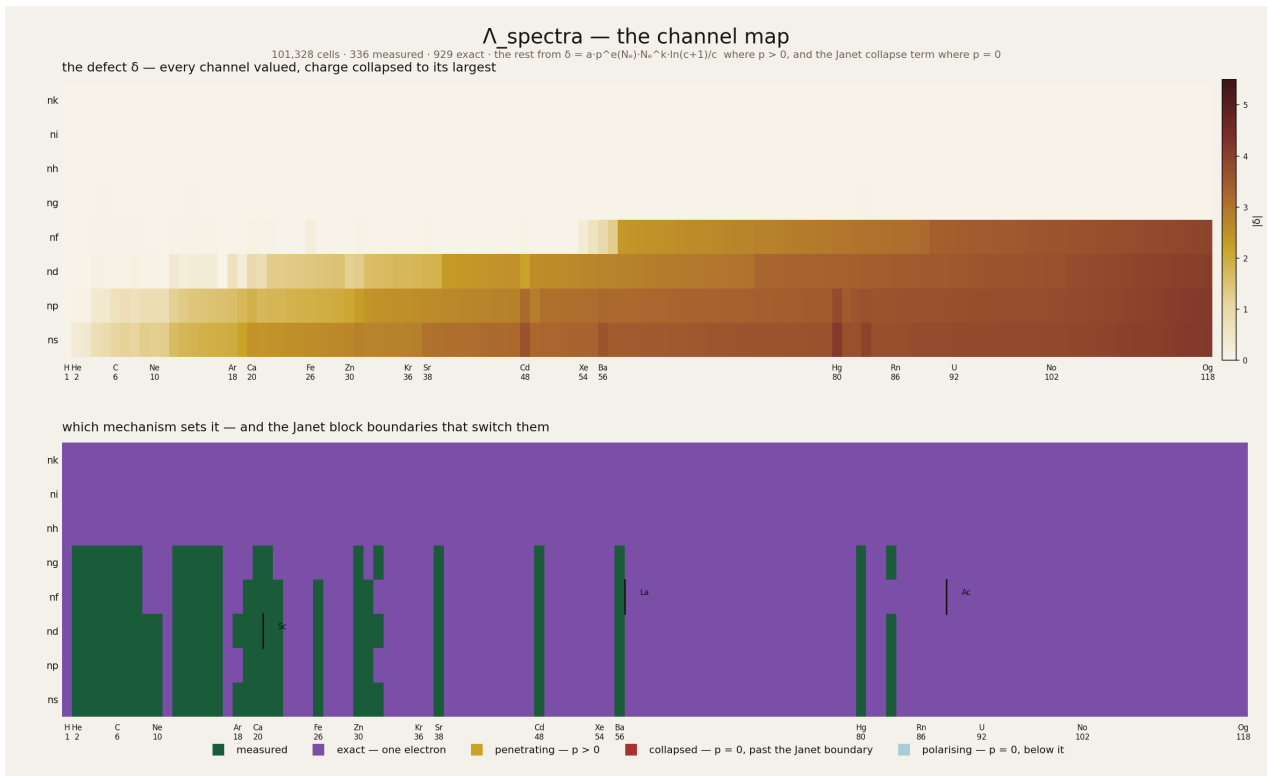
Two channels can both have $p = 0$ — no core orbitals of their own ℓ — and differ by a factor of ten. Ti IV's nd is **0.6202**; Sr II's nf is **0.0618**.

What separates them is **orbital collapse**, and its threshold is a Janet block boundary exactly:

orbital	$n+\ell$	block opens at	element	collapse threshold
3d	5	$Z = 21$	Sc	21
4f	7	$Z = 57$	La	57
5f	8	$Z = 89$	Ac	89

Across 116 cells with $p = 0$ at $\ell = 2$ or 3: **collapsed median 0.637, uncollapsed 0.036**, U-test $p = 9.8 \times 10^{-4}$. The transition is rapid rather than sharp — the largest defects are the atoms *approaching* a boundary: Ca I nd = 0.908 at $Z = 20$ against a threshold of 21, and Ba II nf = 0.756 at 56 against 57.

The collapse coordinate is read from the periodic table, not fitted, and it is why a defect at $\ell = 2$ can be large in one element and negligible in its neighbour.



Every cell valued. The upper panel is δ itself, charge collapsed to its largest; the lower is which mechanism sets it, with the three Janet boundaries marked.

II • THE CHANNELS

Every captured series, with its fit. These are the measured rows of the coordinate table.

Three marks state a limitation. An **asterisk** on the species marks a two-member channel: two points give a defect and one consistency check but cannot detect a perturbation. A **dagger** on the n-range marks an internal gap. And the **limit column** names how the ionisation limit was obtained — *published** where a source quotes it, *theoretical** where it is exact by construction, *constructed** where this work built it by summing two spectra. A channel resting on a limit this work computed is not a measurement against an independent standard, and the row says so.*

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Al I	3s ² nf ² F°	4–55	52	50	50/50	4.0–55.0	+0.0429	0.0552	1	48,278.480
Al I	3s ² nd ² D	3–34	32	30	30/30	2.6–33.7	+0.0248	0.6297	1	48,278.480
Al I	3s ² ns ² S	4–16	13	11	11/11	2.2–14.2	+1.7670	0.0537	1	48,278.480
Al I	3s ² np ² P°	3–7	5	3	3/3	1.5–5.7	+1.3365	0.2080	1	48,278.480
Al II	3sns ¹ S	4–16	13	11	11/11	2.8–14.8	+1.2028	0.0147	2	151,862.500
Al II	3sns ³ S	4–13	10	8	8/8	2.7–11.7	+1.2711	0.0474	2	151,862.500
Al II	3snd ¹ D	3–11	9	7	7/7	3.2–10.9	+0.0603	0.3897	2	151,862.500
Al II	3sng	5–13	9	7	7/7	5.0–13.0	+0.0212	0.0070	2	151,862.500
Al II	3snd ³ D	3–10	8	6	6/6	2.8–9.8	+0.2024	0.0096	2	151,862.500
Al II	3snf ³ F°	4–11	8	6	6/6	3.9–11.0	+0.0140	0.4088	2	151,862.500
Al II	3snp ³ P°	4–10	7	5	5/5	3.1–9.1	+0.8982	0.0571	2	151,862.500
Ar II	(³ P)ns ⁴ P J=5/2	4–6	3	1	1/1	2.2–4.3	+1.7463	0.0477	2	222,848.300
Ar II	(³ P)ns ⁴ P J=3/2	4–6	3	1	1/1	2.2–4.3	+1.7246	0.0663	2	222,848.300
Ar II	(³ P)ns ⁴ P J=1/2	4–6	3	1	1/1	2.2–4.4	+1.6821	0.1507	2	222,848.300

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ar II	(³ P)ns ² P J=3/2	4–6	3	1	1/1	2.3–4.4	+1.6711	0.1084	2	222,848.300
Ar II	(³ P)ns ² P J=1/2	4–6	3	1	1/1	2.3–4.4	+1.6355	0.1526	2	222,848.300
Ar II	(³ P)nd ⁴ D J=7/2	3–5	3	1	1/1	2.2–4.4	+0.6930	0.1695	2	222,848.300
Ar II	(³ P)nd ⁴ D J=5/2	3–5	3	1	1/1	2.2–4.4	+0.6880	0.1753	2	222,848.300
Ar II	(³ P)nd ⁴ D J=3/2	3–5	3	1	1/1	2.2–4.4	+0.6786	0.1919	2	222,848.300
Ar II	(³ P)nd ⁴ D J=1/2	3–5	3	1	1/1	2.2–4.4	+0.6669	0.2155	2	222,848.300
Ar II	(³ P)nd ⁴ F J=9/2	3–5	3	1	1/1	2.3–4.4	+0.6121	0.0885	2	222,848.300
Be I	2snp ¹ P ^o	3–13	11	9	9/9	2.7–12.6	+0.3590	0.1025	1	75,192.500
Be I	2snd ³ D	3–12	10	8	8/8	2.9–11.9	+0.1109	0.0129	1	75,192.500
Be I	2snd ¹ D	3–12	10	8	8/8	3.2–12.1	-0.1152	0.0970	1	75,192.500
Be I	2sns ¹ S	3–11	9	7	7/7	2.3–10.3	+0.6774	0.0136	1	75,192.500
Be I	2sns ³ S	3–8	6	4	4/4	2.2–7.2	+0.7910	0.0429	1	75,192.500
Be I	2snp ³ P ^o	3–8	6	4	4/4	2.6–7.6	+0.3773	0.0374	1	75,192.500
Be I	2snf ³ F ^o	4–7	4	2	2/2	4.0–7.0	+0.0305	0.0063	1	75,192.500
Be II	ns ² S	2–10	9	7	7/7	1.7–9.7	+0.2623	0.0112	2	146,882.860
Be II	np ² P ^o	2–10	9	7	7/7	2.0–10.0	+0.0491	0.0039	2	146,882.860

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Be II	nd ² D	3–9	7	5	5/5	3.0–9.0	+0.0021	0.0004	2	146,882.860
Be II	nf ² F ^o	4–10	7	5	5/5	4.0–10.0	+0.0001	0.0001	2	146,882.860
Be II	ng ² G	5–10	6	4	4/4	5.0–10.0	-0.0001	0.0001	2	146,882.860
Be II	nh ² H ^o	6–10	5	3	3/3	6.0–10.0	-0.0002	0.0001	2	146,882.860
Bi I	(³ P ₀)ns 2[0] J=1/2	7–11	5	3	3/3	2.0–6.1	+4.9036	0.0649	1	58,761.650
Bi I	(³ P ₀)nd 2[2] J=3/2	6–10	5	3	3/3	2.7–6.7	+3.2646	0.0470	1	58,761.650
Bi I	(³ P ₀)nd 2[2] J=5/2	6–9	4	2	2/2	2.8–5.8	+3.2071	0.0272	1	58,761.650
Bi I	(³ P ₀)np 2[1] ^o J=1/2	7–9	3	1	1/1	2.5–4.6	+4.4712	0.0588	1	58,761.650
C II	2s ² ns ² S	3–8	6	4	4/4	2.3–7.3	+0.6624	0.0063	2	196,664.700
C II	2s ² np ² P ^o	3–7	5	3	3/3	2.6–6.7	+0.3928	0.1344	2	196,664.700
C II	2s ² nd ² D	3–7	5	3	3/3	2.9–6.9	+0.0924	0.0645	2	196,664.700
C II	2s ² nf ² F ^o	4–7	4	2	2/2	4.0–7.0	+0.0220	0.0069	2	196,664.700
C II	2s ² ng ² G	5–7	3	1	1/1	5.0–7.0	+0.0054	0.0011	2	196,664.700
Ga I	4s ² ns ² S	5–28	24	22	22/22	2.2–25.2	+2.8013	0.0535	1	48,387.634
Ga I	4s ² nd ² D	4–27	23	21	21/21	2.8–25.7	+1.2926	0.1920	1	48,387.634
Ga I	4s ² np ² P ^o high	41–5	15	13	13/13	38.8–52.7	+2.2107	0.1209	1	48,387.634
Ga I	4s ² nf ² F ^o	4–8	5	3	3/3	4.0–8.0	+0.0252	0.0057	1	48,387.634
Ga I	4s ² np ² P ^o low	4–7	4	2	2/2	1.5–4.7	+2.3450	0.2026	1	48,387.634
He I	1sns ³ S	2–35	25	23	23/23	1.7–34.7	+0.2965	0.0164	1	198,310.666
He I	1sns ¹ S	2–35	25	23	23/23	1.9–34.9	+0.1392	0.0118	1	198,310.666
He I	1snp ¹ P ^o	2–35	25	23	23/23	2.0–35.0	-0.0133	0.0050	1	198,310.666
He I	1snd ¹ D	3–35	24	22	22/22	3.0–35.0	+0.0007	0.0019	1	198,310.666
He I	1snf ¹ F ^o	4–35	23	21	21/21	4.0–35.0	-0.0010	0.0020	1	198,310.666
He I	1snd ³ D	3–35	22	20	20/20	3.0–35.0	+0.0014	0.0018	1	198,310.666
He I	1sng ¹ G	5–35	22	20	20/20	5.0–35.0	-0.0014	0.0020	1	198,310.666
He I	1snh ¹ H ^o	6–35	21	19	19/19	6.0–35.0	-0.0015	0.0020	1	198,310.666
He I	1sni ¹ I	7–35	20	18	18/18	7.0–35.0	-0.0016	0.0019	1	198,310.666
He II	ns	1–10	10	8	8/8	1.0–10.0	-0.0003	0.0006	2	438,908.871
He II	np	2–10	9	7	7/7	2.0–10.0	-0.0004	0.0005	2	438,908.871
He II	nd	3–10	8	6	6/6	3.0–10.0	-0.0004	0.0005	2	438,908.871

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
He II	nf	4–10	7	5	5/5	4.0–10.0	-0.0005	0.0004	2	438,908.871
He II	ng	5–10	6	4	4/4	5.0–10.0	-0.0005	0.0003	2	438,908.871
He II	nh	6–9	4	2	2/2	6.0–9.0	-0.0005	0.0002	2	438,908.871
He II	ni	7–9	3	1	1/1	7.0–9.0	-0.0005	0.0001	2	438,908.871
K II	3p ⁵ nf 2[3/2] J=1	4–6	3	1	1/1	4.0–6.0	+0.0426	0.0069	2	255,072.800
K II	3p ⁵ nf 2[9/2] J=5	4–6	3	1	1/1	4.0–6.0	+0.0336	0.0071	2	255,072.800
K II	3p ⁵ nf 2[5/2] J=3	4–6	3	1	1/1	4.0–6.0	+0.0242	0.0052	2	255,072.800
K II	3p ⁵ nf 2[7/2] J=4	4–6	3	1	1/1	4.0–6.0	+0.0143	0.0059	2	255,072.800
Li I	np 2P ^o	2–42	41	39	39/39	2.0–41.8	+0.1477	0.6779	1	43,487.114
Li I	ns 2S	2–11	10	8	8/8	1.6–10.6	+0.4014	0.0180	1	43,487.114
Li I	nd 2D	3–12	10	8	8/8	3.0–12.0	+0.0031	0.0129	1	43,487.114
Mg II	ns ² S	3–10	8	6	6/6	1.9–8.9	+1.0749	0.0289	2	121,267.640
Mg II	nd ² D	3–10	8	6	6/6	3.0–10.0	+0.0411	0.0145	2	121,267.640
Mg II	np ² P ^o	3–9	7	5	5/5	2.3–8.3	+0.7077	0.0361	2	121,267.640
Mg II	nf ² F ^o	4–10	7	5	5/5	4.0–10.0	+0.0030	0.0009	2	121,267.640
Mg II	ng ² G	5–10	6	4	4/4	5.0–10.0	+0.0007	0.0001	2	121,267.640
Mg II	nh ² H ^o	6–9	4	2	2/2	6.0–9.0	+0.0002	0.0000	2	121,267.640
Mg II	ni ² I	7–9	3	1	1/1	7.0–9.0	+0.0000	0.0000	2	121,267.640
N II	2p-ns ³ P ^o J=0	3–6	4	2	2/2	2.2–5.2	+0.7747	0.0264	2	238,750.500
N II	2p-ns ³ P ^o J=1	3–6	4	2	2/2	2.2–5.2	+0.7731	0.0260	2	238,750.500
N II	2p-ns ³ P ^o J=2	3–6	4	2	2/2	2.2–5.3	+0.7633	0.0464	2	238,750.500
N II	2p-ns ¹ P ^o J=1	3–6	4	2	2/2	2.2–5.3	+0.7325	0.0824	2	238,750.500
Na I	ns	3–20	18	16	16/16	1.6–18.7	+1.3506	0.0250	1	41,449.451
Na I	np	3–20	18	16	16/16	2.1–19.1	+0.8584	0.0279	1	41,449.451
Na I	nd	3–20	18	16	16/16	3.0–20.0	+0.0141	0.0045	1	41,449.451
Na I	nf	4–20	17	15	15/15	4.0–20.0	+0.0014	0.0004	1	41,449.451
Na I	ng	5–20	16	14	14/14	5.0–20.0	+0.0003	0.0001	1	41,449.451
Na I	nh	6–10	5	3	3/3	6.0–10.0	+0.0000	0.0000	1	41,449.451
Na II	(² P ^o _{3/2})ns 2[3/2] ^o J=2	3–6	4	2	2/2	1.9–5.0	+1.0385	0.0284	2	381,390.200
Na II	(² P ^o _{3/2})ns 2[3/2] ^o J=1	3–6	4	2	2/2	1.9–5.0	+1.0268	0.0371	2	381,390.200
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Na II	(² P ^o _{3/2})nd 2[1/2] ^o J=0	3–5	3	1	1/1	2.9–4.9	+0.0671	0.0083	2	381,390.200
Na II	(² P ^o _{3/2})nd 2[1/2] ^o J=1	3–5	3	1	1/1	2.9–4.9	+0.0641	0.0085	2	381,390.200
Na II	(² P ^o _{3/2})nd 2[3/2] ^o J=2	3–5	3	1	1/1	2.9–4.9	+0.0556	0.0027	2	381,390.200
Na II	(² P ^o _{3/2})nd 2[3/2] ^o J=1	3–5	3	1	1/1	3.0–5.0	+0.0268	0.0026	2	381,390.200
Zn II	ns ² S	4–9	6	4	4/4	1.7–6.8	+2.2084	0.0679	2	144,892.600
Zn II	nd ² D	4–9	6	4	4/4	3.0–8.0	+0.9583	0.0228	2	144,892.600
Zn II	np ² P ^o	4–8	5	3	3/3	2.1–6.1	+1.8386	0.0616	2	144,892.600
Zn II	nf ² F ^o	4–8	5	3	3/3	4.0–8.0	+0.0172	0.0044	2	144,892.600
Zn II	ng ² G	5–7	3	1	1/1	5.0–7.0	+0.0038	0.0008	2	144,892.600
Bi III	6s2.nd 2D J=3/2	6–8	3	1	untested	3.0–5.2	+2.8926	0.0795	3	206,242.000
Bi III	6s2.nd 2D J=5/2	6–8	3	1	untested	3.1–5.2	+2.8456	0.0496	3	206,242.000
Bi III	6s2.ng 2G J=7/2	5–8	4	2	untested	5.0–8.0	+0.0378	0.0043	3	206,242.000
Bi III	6s2.ng 2G J=9/2	5–8	4	2	untested	5.0–8.0	+0.0383	0.0046	3	206,242.000
Bi III	6s2.np 2P* J=1/2	6–8	3	1	untested	2.2–4.4	+3.7096	0.0732	3	206,242.000
Bi III	6s2.np 2P* J=3/2	6–8	3	1	untested	2.3–4.5	+3.6044	0.0630	3	206,242.000
Bi III	6s2.ns 2S J=1/2	7–9	3	1	untested	3.0–5.0	+3.9869	0.0234	3	206,242.000

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
C II	2s2(1S)np 2P° J=1/2	2–4	3	1	untested	1.5–3.6	+0.4403	0.0469	2	196,664.700
C II	2s2(1S)np 2P° J=3/2	2–4	3	1	untested	1.5–3.6	+0.4401	0.0469	2	196,664.700
Li I	np 2P° J=1/2	2–4	3	1	untested	2.0–4.0	+0.0436	0.0021	1	43,487.150
Li I	np 2P° J=3/2	2–4	3	1	untested	2.0–4.0	+0.0436	0.0021	1	43,487.150
Mg I	3snd 1D J=2	3–5	3	1	untested	2.7–4.5	+0.4034	0.0647	1	61,671.050
Mg I	3snd 3D J=1	3–5	3	1	untested	2.8–4.8	+0.1701	0.0009	1	61,671.050
Mg I	3snd 3D J=2	3–5	3	1	untested	2.8–4.8	+0.1701	0.0008	1	61,671.050
Mg I	3snd 3D J=3	3–5	3	1	untested	2.8–4.8	+0.1701	0.0008	1	61,671.050
Mg I	3snp 1P° J=1	3–6	4	2	untested	2.0–5.0	+1.0135	0.0265	1	61,671.050
Mg I	3snp 3P° J=0	3–6	4	2	untested	1.7–4.9	+1.2056	0.0789	1	61,671.050
Mg I	3snp 3P° J=1	3–6	4	2	untested	1.7–4.9	+1.2052	0.0789	1	61,671.050
Mg I	3snp 3P° J=2	3–6	4	2	untested	1.7–4.9	+1.2045	0.0788	1	61,671.050
Mg I	3sns 1S J=0	4–6	3	1	untested	2.5–4.5	+1.5331	0.0067	1	61,671.050
Mg I	3sns 3S J=1	4–6	3	1	untested	2.3–4.4	+1.6603	0.0180	1	61,671.050
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Mg II	np 2P° J=1/2	3–5	3	1	untested	2.3–4.3	+0.7187	0.0123	2	121,267.640
Mg II	np 2P° J=3/2	3–5	3	1	untested	2.3–4.3	+0.7174	0.0123	2	121,267.640
Mg II	ns 2S J=1/2	3–6	4	2	untested	1.9–4.9	+1.0806	0.0102	2	121,267.640
Sc III	3p6.nd 2D J=3/2	3–7	5	3	untested	2.2–6.4	+0.6538	0.0617	3	199,677.370
Sc III	3p6.nd 2D J=5/2	3–7	5	3	untested	2.2–6.4	+0.6529	0.0616	3	199,677.370
Sc III	3p6.nf 2F* J=5/2	4–7	4	2	untested	4.0–6.9	+0.0450	0.0069	3	199,677.370
Sc III	3p6.nf 2F* J=7/2	4–7	4	2	untested	4.0–6.9	+0.0450	0.0069	3	199,677.370
Sc III	3p6.ng 2G J=7/2	5–8	4	2	untested	5.0–8.0	+0.0073	0.0007	3	199,677.370
Sc III	3p6.ng 2G J=9/2	5–8	4	2	untested	5.0–8.0	+0.0073	0.0007	3	199,677.370
Sc III	3p6.np 2P* J=1/2	4–7	4	2	untested	2.7–5.7	+1.2861	0.0211	3	199,677.370
Sc III	3p6.np 2P* J=3/2	4–7	4	2	untested	2.7–5.7	+1.2815	0.0211	3	199,677.370
Sc III	3p6.ns 2S J=1/2	4–8	5	3	untested	2.4–6.4	+1.5848	0.0180	3	199,677.370
Si II	3s2.nd 2D J=3/2	3–8	6	4	untested	2.9–7.7	+0.2290	0.0556	2	131,838.140
Si II	3s2.nd 2D J=5/2	3–8	6	4	untested	2.9–7.7	+0.2289	0.0558	2	131,838.140
Si II	3s2.nf 2F* J=5/2	4–11	8	6	untested	3.9–10.9	+0.0846	0.0120	2	131,838.140
Si II	3s2.nf 2F* J=7/2	4–11	8	6	untested	3.9–10.9	+0.0846	0.0120	2	131,838.140
Si II	3s2.ng 2G J=7/2	5–10	6	4	untested	5.0–10.0	+0.0174	0.0015	2	131,838.140
Si II	3s2.ng 2G J=9/2	5–10	6	4	untested	5.0–10.0	+0.0174	0.0015	2	131,838.140
Si II	3s2.np 2P* J=1/2	3–10	8	6	untested	1.8–9.1	+1.0239	0.0878	2	131,838.140
Si II	3s2.np 2P* J=3/2	3–10	8	6	untested	1.8–9.1	+1.0215	0.0868	2	131,838.140
Si II	3s2.ns 2S J=1/2	4–9	6	4	untested	2.6–7.6	+1.3944	0.0161	2	131,838.140
Li III	nd 2D J=3/2	3–9	7	5	untested	3.0–9.0	+0.0003	0.0002	3	987,662.290
Li III	nd 2D J=5/2	3–9	7	5	untested	3.0–9.0	+0.0002	0.0002	3	987,662.290
Li III	nf 2F* J=5/2	4–9	6	4	untested	4.0–9.0	+0.0003	0.0002	3	987,662.290
Li III	nf 2F* J=7/2	4–9	6	4	untested	4.0–9.0	+0.0002	0.0002	3	987,662.290
Li III	ng 2G J=7/2	5–9	5	3	untested	5.0–9.0	+0.0003	0.0001	3	987,662.290
Li III	ng 2G J=9/2	5–9	5	3	untested	5.0–9.0	+0.0003	0.0001	3	987,662.290
Li III	nh 2H* J=11/2	6–9	4	2	untested	6.0–9.0	+0.0003	0.0001	3	987,662.290
Li III	nh 2H* J=9/2	6–9	4	2	untested	6.0–9.0	+0.0003	0.0001	3	987,662.290
Li III	ni 2I J=11/2	7–9	3	1	untested	7.0–9.0	+0.0004	0.0001	3	987,662.290
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Li III	ni 2I J=13/2	7–9	3	1	untested	7.0–9.0	+0.0004	0.0001	3	987,662.290

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Li III	np 2P* J=1/2	2–10	9	7	untested	2.0–10.0	+0.0004	0.0002	3	987,662.290
Li III	np 2P* J=3/2	2–10	9	7	untested	2.0–10.0	+0.0003	0.0002	3	987,662.290
Li III	ns 2S J=1/2	1–9	9	7	untested	1.0–9.0	+0.0003	0.0002	3	987,662.290
Ba III	5p5.(2P*<3/2>).nd 2[1/2]* J=0	5–7	3	1	untested	2.5–4.7	+2.3509	0.0972	3	289,100.000
Ba III	5p5.(2P*<3/2>).nd 2[1/2]* J=1	5–7	3	1	untested	2.5–4.8	+2.3360	0.0975	3	289,100.000
Ba III	5p5.(2P*<3/2>).nd 2[3/2]* J=1	5–7	3	1	untested	2.7–4.9	+2.1891	0.0829	3	289,100.000
Ba III	5p5.(2P*<3/2>).nd 2[3/2]* J=2	5–7	3	1	untested	2.6–4.8	+2.3061	0.0981	3	289,100.000
Ba III	5p5.(2P*<3/2>).nd 2[5/2]* J=2	5–7	3	1	untested	2.6–4.8	+2.2821	0.0849	3	289,100.000
Ba III	5p5.(2P*<3/2>).nd 2[5/2]* J=3	5–7	3	1	untested	2.6–4.8	+2.2601	0.0738	3	289,100.000
Ba III	5p5.(2P*<3/2>).nd 2[7/2]* J=3	5–7	3	1	untested	2.6–4.8	+2.3041	0.0846	3	289,100.000
Ba III	5p5.(2P*<3/2>).nd 2[7/2]* J=4	5–7	3	1	untested	2.6–4.8	+2.3113	0.0890	3	289,100.000
Ba III	5p5.(2P*<3/2>).ns 2[3/2]* J=1	6–8	3	1	untested	2.7–4.8	+3.2585	0.0242	3	289,100.000
Ba III	5p5.(2P*<3/2>).ns 2[3/2]* J=2	6–8	3	1	untested	2.7–4.7	+3.2795	0.0267	3	289,100.000
Bi II	6s2.6p.nd (1/2,3/2)* J=1	6–14	9	7	untested	2.8–11.0	+2.9846	0.0626	2	134,720.000
Bi II	6s2.6p.nd (1/2,3/2)* J=2	6–10†	4	2	untested	2.8–6.9	+3.1079	0.0545	2	134,720.000
Bi II	6s2.6p.nd (1/2,5/2)* J=2	6–8	3	1	untested	2.9–5.0	+3.0001	0.1072	2	134,720.000
Bi II	6s2.6p.nf (1/2,5/2) J=3	5–8†	3	1	untested	3.9–6.8	+1.1755	0.0437	2	134,720.000
Bi II	6s2.6p.nf (1/2,7/2) J=4	5–8†	3	1	untested	3.9–6.8	+1.1468	0.0343	2	134,720.000
Bi II	6s2.6p.ns (1/2,1/2)* J=0	7–9	3	1	untested	2.6–4.7	+4.3705	0.0333	2	134,720.000
Bi II	6s2.6p.ns (1/2,1/2)* J=1	7–9	3	1	untested	2.6–4.7	+4.3684	0.0313	2	134,720.000
Ne I	2s22p5(2P*3/2)nd 2[3/2]* J=1	13–20	8	6	untested	13.0–20.0	+0.0154	0.0024	1	173,929.750
Ne I	2s22p5(2P*3/2)ns 2[3/2]* J=1	12–20	9	7	untested	10.7–18.7	+1.2923	0.0047	1	173,929.750
Ne I	nd 2[3/2]* J=1	11–20	10	8	untested	11.0–20.0	+0.0119	0.0043	1	174,710.090
Ne I	ns 2[1/2]* J=1	11–20	10	8	untested	9.7–18.7	+1.2974	0.0038	1	174,710.090
Li II	1s.nd 1D J=2	3–10	8	6	untested	3.0–10.0	+0.0011	0.0001	2	610,078.400
Li II	1s.nd 3D J=2	3–10	8	6	untested	3.0–10.0	+0.0027	0.0003	2	610,078.400
Li II	1s.nf 1F* J=3	4–10	7	5	untested	4.0–10.0	+0.0000	0.0002	2	610,078.400
Li II	1s.nf 3F* J=3	4–10	7	5	untested	4.0–10.0	+0.0006	0.0002	2	610,078.400
Li II	1s.ng 1G J=4	5–9	5	3	untested	5.0–9.0	+0.0000	0.0000	2	610,078.400
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Li II	1s.nh 1H* J=5	6–9	4	2	untested	6.0–9.0	-0.0000	0.0000	2	610,078.400
Li II	1s.np 1P* J=1	2–14	13	11	untested	2.0–14.0	-0.0144	0.0014	2	610,078.400
Li II	1s.np 3P* J=1	2–10	9	7	untested	1.9–9.9	+0.0543	0.0004	2	610,078.400
Li II	1s.ns 1S J=0	2–10	9	7	untested	1.9–9.9	+0.0744	0.0010	2	610,078.400
Li II	1s.ns 3S J=1	2–10	9	7	untested	1.8–9.8	+0.1814	0.0035	2	610,078.400
Si I	nd (3/2,3/2)* J=1	20–50	31	29	29/29	20.0–50.0	+0.0126	0.0444	1	66,035.000

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Si I	nd (3/2,5/2)* J=3	20–56	37	35	35/35	19.9–56.0	+0.0570	0.0366	1	66,035.000
Zn I	nd 1D J=2	12–20	9	7	untested	10.8–18.8	+1.2232	0.0018	1	75,769.310
Zn I	np 1P* J=1	13–40	28	26	untested	10.9–37.9	+2.0966	0.0033	1	75,769.310
Ba III	nd 2[3/2]* J=2	7–16	10	8	8/8	4.8–13.8	+2.1786	0.0136	3	306,650.000
Ba III	ns 2[1/2]* J=1	8–17	10	8	untested	4.8–13.8	+3.2300	0.0154	3	306,650.000
Ba III	nd 2[3/2]* J=2	8–22†	14	12	untested	5.9–19.9	+2.1273	0.0227	3	289,100.000
Ba III	ns 2[3/2]* J=1	9–23	15	13	untested	5.8–19.9	+3.2018	0.0273	3	289,100.000
Ne II	nd 4D J=3/2	3–6	4	2	untested	2.9–5.9	+0.0700	0.0012	2	330,388.600
Ne II	nd 4D J=5/2	3–6	4	2	untested	2.9–5.9	+0.0773	0.0036	2	330,388.600
Ne II	nd 4D J=7/2	3–6	4	2	untested	2.9–5.9	+0.0813	0.0047	2	330,388.600
Ne II	nd 4F J=9/2	3–6	4	2	untested	3.0–5.9	+0.0547	0.0070	2	330,388.600
Ne II	ns 2P J=1/2	3–7	5	3	untested	2.0–6.3	+0.8636	0.0857	2	330,388.600
Ne II	ns 2P J=3/2	3–7	5	3	untested	2.0–6.0	+0.9480	0.0143	2	330,388.600
Ne II	ns 4P J=1/2	3–7	5	3	untested	2.0–6.2	+0.9132	0.0743	2	330,388.600
Ne II	ns 4P J=3/2	3–7	5	3	untested	2.0–6.2	+0.9261	0.0730	2	330,388.600
Ne II	ns 4P J=5/2	3–7	5	3	untested	2.0–6.0	+0.9906	0.0125	2	330,388.600
Ne II	nd 2S J=1/2	3–8	6	4	untested	3.0–7.9	+0.0701	0.0422	2	356,229.300
Ne II	ns 2D J=3/2	3–5	3	1	untested	2.0–4.0	+0.9850	0.0115	2	356,229.300
Ne II	ns 2D J=5/2	3–5	3	1	untested	2.0–4.0	+0.9850	0.0115	2	356,229.300
Ne I	np 1P* J=1	3–12	10	8	untested	2.2–11.1	+0.8409	0.0191	1	390,977.350
Ne II	np 2D* J=3/2	3–6	4	2	untested	2.4–5.5	+0.5651	0.0645	2	330,388.600
Ne II	np 2D* J=5/2	3–6	4	2	untested	2.4–5.4	+0.6294	0.0117	2	330,388.600
Ne II	np 2S* J=1/2	3–6	4	2	untested	2.4–5.4	+0.6010	0.0199	2	330,388.600
Ne II	np 4D* J=1/2	3–6	4	2	untested	2.3–5.5	+0.5916	0.0576	2	330,388.600
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ne II	np 4D* J=5/2	3–6	4	2	untested	2.3–5.5	+0.6055	0.0551	2	330,388.600
Ne II	np 4D* J=7/2	3–6	4	2	untested	2.3–5.4	+0.6541	0.0133	2	330,388.600
Ne II	np 4P* J=1/2	3–6	4	2	untested	2.3–5.4	+0.6430	0.0585	2	330,388.600
Ne II	np 4P* J=3/2	3–6	4	2	untested	2.3–5.3	+0.6767	0.0235	2	330,388.600
Ne II	np 4P* J=5/2	3–6	4	2	untested	2.3–5.3	+0.6884	0.0174	2	330,388.600
Ne II	np 4S* J=3/2	3–6	4	2	untested	2.4–5.5	+0.5679	0.0424	2	330,388.600
Ne II	nf 2[2]* J=5/2	5–8	4	2	untested	5.0–8.0	+0.0039	0.0001	2	330,388.600
Ne II	nf 2[3]* J=7/2	5–8	4	2	untested	5.0–8.0	+0.0086	0.0001	2	330,388.600
Ne II	nf 2[4]* J=7/2	5–8	4	2	untested	5.0–8.0	+0.0096	0.0002	2	330,388.600
Ne II	nf 2[4]* J=9/2	5–8	4	2	untested	5.0–8.0	+0.0098	0.0002	2	330,388.600
Ne II	nf 2[5]* J=11/2	5–8	4	2	untested	5.0–8.0	+0.0025	0.0003	2	330,388.600
Ne II	ng 2[2] J=3/2	5–7	3	1	untested	5.0–7.0	-0.0016	0.0001	2	330,388.600
Ne II	ng 2[3] J=5/2	5–7	3	1	untested	5.0–7.0	+0.0010	0.0001	2	330,388.600
Ne II	ng 2[4] J=7/2	5–7	3	1	untested	5.0–7.0	+0.0029	0.0001	2	330,388.600
Ne II	ng 2[5] J=9/2	5–7	3	1	untested	5.0–7.0	+0.0030	0.0001	2	330,388.600
Ne II	ng 2[6] J=11/2	5–7	3	1	untested	5.0–7.0	-0.0002	0.0001	2	330,388.600
Ar II	ng 2[4] J=9/2	6–8	3	1	untested	6.0–8.0	+0.0089	0.0036	2	222,848.300
Ar II	ng 2[6] J=11/2	6–8	3	1	untested	6.0–8.0	+0.0035	0.0003	2	222,848.300
Ar II	ng 2[6] J=13/2	6–8	3	1	untested	6.0–8.0	+0.0035	0.0002	2	222,848.300
Ar II *	ng 2[2] J=3/2	6–7	2	1	untested	6.0–7.0	-0.0006	0.0002	2	222,848.300
Ar II *	ng 2[2] J=5/2	6–7	2	1	untested	6.0–7.0	-0.0006	0.0002	2	222,848.300
Ar II *	ng 2[3] J=5/2	6–7	2	1	untested	6.0–7.0	+0.0063	0.0001	2	222,848.300

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ar II *	ng 2[3] J=7/2	6-7	2	1	untested	6.0-7.0	+0.0063	0.0001	2	222,848.300
Ar II *	ng 2[4] J=7/2	6-7	2	1	untested	6.0-7.0	+0.0114	0.0001	2	222,848.300
Ar II *	ng 2[5] J=11/2	6-7	2	1	untested	6.0-7.0	+0.0119	0.0002	2	222,848.300
Ar II *	ng 2[5] J=9/2	6-7	2	1	untested	6.0-7.0	+0.0119	0.0002	2	222,848.300
Ar II *	nh 2[3]* J=5/2	6-7	2	1	untested	6.0-7.0	-0.0017	0.0001	2	222,848.300
Ar II *	nh 2[3]* J=7/2	6-7	2	1	untested	6.0-7.0	-0.0017	0.0001	2	222,848.300
Ar II *	nh 2[4]* J=7/2	6-7	2	1	untested	6.0-7.0	+0.0021	0.0001	2	222,848.300
Ar II *	nh 2[4]* J=9/2	6-7	2	1	untested	6.0-7.0	+0.0021	0.0001	2	222,848.300
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ar II *	nh 2[5]* J=11/2	6-7	2	1	untested	6.0-7.0	+0.0047	0.0001	2	222,848.300
Ar II *	nh 2[5]* J=9/2	6-7	2	1	untested	6.0-7.0	+0.0047	0.0001	2	222,848.300
Ar II *	nh 2[6]* J=11/2	6-7	2	1	untested	6.0-7.0	+0.0045	0.0001	2	222,848.300
Ar II *	nh 2[6]* J=13/2	6-7	2	1	untested	6.0-7.0	+0.0045	0.0001	2	222,848.300
Ar II *	nh 2[7]* J=13/2	6-7	2	1	untested	6.0-7.0	-0.0000	0.0001	2	222,848.300
Ar II *	nh 2[7]* J=15/2	6-7	2	1	untested	6.0-7.0	-0.0000	0.0001	2	222,848.300
Ar II *	ni 2[4] J=7/2	7-8	2	1	untested	7.0-8.0	-0.0009	0.0005	2	222,848.300
Ar II *	ni 2[4] J=9/2	7-8	2	1	untested	7.0-8.0	-0.0009	0.0005	2	222,848.300
Ar II *	ni 2[5] J=11/2	7-8	2	1	untested	7.0-8.0	+0.0012	0.0003	2	222,848.300
Ar II *	ni 2[5] J=9/2	7-8	2	1	untested	7.0-8.0	+0.0012	0.0003	2	222,848.300
Ar II *	ni 2[6] J=11/2	7-8	2	1	untested	7.0-8.0	+0.0025	0.0002	2	222,848.300
Ar II *	ni 2[6] J=13/2	7-8	2	1	untested	7.0-8.0	+0.0025	0.0002	2	222,848.300
Ar II *	ni 2[7] J=13/2	7-8	2	1	untested	7.0-8.0	+0.0023	0.0002	2	222,848.300
Ar II *	ni 2[7] J=15/2	7-8	2	1	untested	7.0-8.0	+0.0023	0.0002	2	222,848.300
Ar II *	ni 2[8] J=15/2	7-8	2	1	untested	7.0-8.0	-0.0000	0.0006	2	222,848.300
Ar II *	ni 2[8] J=17/2	7-8	2	1	untested	7.0-8.0	-0.0000	0.0006	2	222,848.300
Bi II *	6s2.6p.nf (1/2,7/2) J=3	5-6	2	1	untested	3.9-4.9	+1.1419	0.0039	2	134,720.000
Bi II *	6s2.6p.np (1/2,1/2) J=0	7-8	2	1	untested	3.0-4.1	+3.9567	0.0079	2	134,720.000
Bi III *	6s2.nh 2H* J=11/2	6-7	2	1	untested	6.0-7.0	+0.0103	0.0005	3	206,242.000
Bi III *	6s2.nh 2H* J=9/2	6-7	2	1	untested	6.0-7.0	+0.0103	0.0005	3	206,242.000
C II *	2s2(1S)nd 2D J=3/2	3-4	2	1	untested	2.9-3.9	+0.0740	0.0044	2	196,664.700
C II *	2s2(1S)nd 2D J=5/2	3-4	2	1	untested	2.9-3.9	+0.0740	0.0044	2	196,664.700
C II *	2s2(1S)ns 2S J=1/2	3-4	2	1	untested	2.3-3.3	+0.6615	0.0020	2	196,664.700
Li I *	nd 2D J=3/2	3-4	2	1	untested	3.0-4.0	+0.0016	0.0001	1	43,487.150
Li I *	nd 2D J=5/2	3-4	2	1	untested	3.0-4.0	+0.0016	0.0001	1	43,487.150
Li I *	ns 2S J=1/2	2-3	2	1	untested	1.6-2.6	+0.4077	0.0038	1	43,487.150
Li II *	1s.ni 1I J=6	7-8	2	1	untested	7.0-8.0	-0.0001	0.0000	2	610,078.400
Li II *	1s.np 3P* J=0	2-3	2	1	untested	1.9-2.9	+0.0537	0.0004	2	610,078.400
Li II *	1s.np 3P* J=2	2-3	2	1	untested	1.9-2.9	+0.0537	0.0004	2	610,078.400
Li III *	nk 2K* J=13/2	8-9	2	1	untested	8.0-9.0	+0.0004	0.0001	3	987,662.290
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Li III *	nk 2K* J=15/2	8-9	2	1	untested	8.0-9.0	+0.0004	0.0001	3	987,662.290
Mg I *	3snf 1F° J=3	4-5	2	1	untested	4.0-5.0	+0.0413	0.0022	1	61,671.050
Mg I *	3snf 3F° J=2	4-5	2	1	untested	4.0-5.0	+0.0413	0.0022	1	61,671.050
Mg I *	3snf 3F° J=3	4-5	2	1	untested	4.0-5.0	+0.0413	0.0022	1	61,671.050
Mg I *	3snf 3F° J=4	4-5	2	1	untested	4.0-5.0	+0.0413	0.0022	1	61,671.050
Mg II *	nd 2D J=3/2	3-4	2	1	untested	3.0-4.0	+0.0340	0.0036	2	121,267.640

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Mg II *	nd 2D J=5/2	3–4	2	1	untested	3.0–4.0	+0.0340	0.0036	2	121,267.640
Ne I *	2p5(2P°3/2)nd 2[7/2] ^o J=4	3–4	2	1	untested	3.0–4.0	+0.0193	0.0014	1	173,929.750
Ne I *	2p5(2P°3/2)ns 2[3/2] ^o J=2	3–4	2	1	untested	1.7–2.7	+1.3329	0.0085	1	173,929.750
Ne II *	nd 2F J=5/2	3–4	2	1	untested	3.0–4.0	+0.0289	0.0055	2	356,229.300
Ne II *	nd 2F J=7/2	3–4	2	1	untested	3.0–4.0	+0.0289	0.0055	2	356,229.300
Ne II *	nd 2G J=7/2	3–4	2	1	untested	2.9–3.9	+0.0674	0.0050	2	356,229.300
Ne II *	nd 2G J=9/2	3–4	2	1	untested	2.9–3.9	+0.0674	0.0050	2	356,229.300
Ne II *	nd 2P J=3/2	3–4	2	1	untested	2.9–3.9	+0.0605	0.0040	2	356,229.300
Sc III *	3p6.nh 2H* J=11/2	6–7	2	1	untested	6.0–7.0	+0.0022	0.0001	3	199,677.370
Sc III *	3p6.nh 2H* J=9/2	6–7	2	1	untested	6.0–7.0	+0.0022	0.0001	3	199,677.370
Zn I *	4snd 3D J=1	4–5	2	1	untested	2.9–3.9	+1.0940	0.0007	1	75,769.330
Zn I *	4snd 3D J=2	4–5	2	1	untested	2.9–3.9	+1.0936	0.0007	1	75,769.330
Zn I *	4snd 3D J=3	4–5	2	1	untested	2.9–3.9	+1.0930	0.0007	1	75,769.330
Ba III *	ng 2[11/2]* J=5	5–6	2	1	untested	5.0–6.0	+0.0394	0.0028	3	289,100.000
Ba III *	ng 2[11/2]* J=6	5–6	2	1	untested	5.0–6.0	+0.0397	0.0029	3	289,100.000
Ba III *	ng 2[5/2]* J=2	5–6	2	1	untested	5.0–6.0	+0.0428	0.0002	3	289,100.000
Ba III *	ng 2[5/2]* J=3	5–6	2	1	untested	5.0–6.0	+0.0422	0.0000	3	289,100.000
Ba III *	ng 2[7/2]* J=3	5–6	2	1	untested	5.0–6.0	+0.0208	0.0006	3	289,100.000
Ba III *	ng 2[7/2]* J=4	5–6	2	1	untested	5.0–6.0	+0.0212	0.0007	3	289,100.000
Ba III *	ng 2[9/2]* J=4	5–6	2	1	untested	5.0–6.0	+0.0174	0.0030	3	289,100.000
Ba III *	ng 2[9/2]* J=5	5–6	2	1	untested	5.0–6.0	+0.0172	0.0029	3	289,100.000
Si I	ns (3/2,1/2)* J=1	11–24	14	12	untested	9.1–22.1	+1.8546	0.0082	1	66,035.000
Si I	ns (3/2,1/2)* J=2	11–19	9	7	untested	9.1–17.1	+1.8883	0.0005	1	66,035.000
Si I	nf 2[3/2] J=1	4–7	4	2	untested	4.0–7.0	-0.0003	0.0026	1	66,035.000
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Si I	nf 2[3/2] J=2	4–7	4	2	untested	4.0–7.0	-0.0008	0.0023	1	66,035.000
Si I	nf 2[5/2] J=2	4–7	4	2	untested	4.0–7.0	+0.0277	0.0056	1	66,035.000
Si I	nf 2[5/2] J=3	4–7	4	2	untested	4.0–7.0	+0.0281	0.0057	1	66,035.000
Si I	nf 2[7/2] J=3	4–7	4	2	untested	4.0–6.9	+0.0471	0.0041	1	66,035.000
Si I	nf 2[7/2] J=4	4–7	4	2	untested	4.0–6.9	+0.0462	0.0039	1	66,035.000
Si I	nf 2[9/2] J=4	4–7	4	2	untested	4.0–7.0	+0.0165	0.0023	1	66,035.000
Si I	nf 2[9/2] J=5	4–7	4	2	untested	4.0–7.0	+0.0181	0.0027	1	66,035.000
Si I	nd 1D* J=2	3–8	6	4	untested	2.4–7.9	+0.4032	0.1771	1	65,747.760
Si I	nd 1F* J=3	3–8†	5	3	untested	3.0–8.4	-0.1375	0.1513	1	65,747.760
Si I	nd 1P* J=1	3–8	6	4	untested	3.0–8.0	+0.0284	0.0044	1	65,747.760
Si I	nd 3F* J=2	3–8	6	4	untested	2.6–7.6	+0.3730	0.0333	1	65,747.760
Si I	nd 3F* J=3	3–8	6	4	untested	2.6–7.8	+0.3007	0.0615	1	65,747.760
Si I	nd 3F* J=4	3–8	6	4	untested	2.6–8.2	+0.1429	0.2129	1	65,747.760
Si I *	np (1/2,1/2) J=1	6–7	2	1	untested	4.5–5.5	+1.4808	0.0027	1	65,747.760
Si I *	np (1/2,3/2) J=1	6–7	2	1	untested	4.6–5.6	+1.4252	0.0076	1	65,747.760
Si I *	np (1/2,3/2) J=2	6–7	2	1	untested	4.6–5.6	+1.4210	0.0078	1	65,747.760
Si I	ns (1/2,1/2)* J=0	6–10	5	3	3/3	4.1–8.1	+1.8930	0.0038	1	65,747.760
Si I	ns (1/2,1/2)* J=1	6–10	5	3	3/3	4.1–8.1	+1.8758	0.0091	1	65,747.760
Si I	ns (1/2,1/2)* J=0	13–21	9	7	untested	11.1–19.1	+1.8873	0.0031	1	65,747.760
Si I	ns (1/2,1/2)* J=1	13–21	9	7	untested	11.1–19.1	+1.8765	0.0105	1	65,747.760
Si I	nd (1/2,3/2)* J=1	14–44	29	27	27/27	14.0–43.9	+0.0612	0.0310	1	65,747.760

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Si I	nd (1/2,3/2)* J=1	9–44†	34	32	untested	9.0–43.9	+0.0584	0.0312	1	65,747.760
Si I	nd (3/2,3/2)* J=0	8–13	6	4	4/4	8.1–13.1	-0.0680	0.0129	1	66,035.000
Si I	nd (3/2,3/2)* J=1	20–50	31	29	untested	20.0–50.0	+0.0129	0.0444	1	66,035.000
Si I	nd (3/2,5/2)* J=1	8–13	6	4	4/4	7.9–12.9	+0.0784	0.0064	1	66,035.000
Si I	nd (3/2,5/2)* J=3	20–56	37	35	untested	19.9–56.0	+0.0573	0.0366	1	66,035.000
Si I	nd (3/2,3/2)* J=0	8–16	9	7	untested	8.1–16.0	-0.0609	0.0146	1	66,035.000
Si I	nd (3/2,5/2)* J=1	8–16	9	7	untested	7.9–15.9	+0.0816	0.0073	1	66,035.000
B III	nd 2D J=3/2	3–11	9	7	untested	3.0–11.0	+0.0023	0.0002	3	305,930.800
B III	nf 2F* J=5/2	4–11	8	6	untested	4.0–11.0	+0.0003	0.0000	3	305,930.800
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
B III	ng 2G J=7/2	5–10	6	4	untested	5.0–10.0	+0.0001	0.0000	3	305,930.800
B III	nh 2H* J=9/2	6–10	5	3	untested	6.0–10.0	+0.0000	0.0000	3	305,930.800
B III	np 2P* J=1/2	2–9	8	6	untested	2.0–9.0	+0.0437	0.0007	3	305,930.800
B III	ns 2S J=1/2	2–9	8	6	untested	1.8–8.8	+0.1959	0.0030	3	305,930.800
Be III	ls.nd 1D J=2	3–10	8	6	untested	3.0–10.0	+0.0008	0.0002	3	1,241,256.601
Be III	ls.nd 3D J=2	3–10	8	6	untested	3.0–10.0	+0.0023	0.0002	3	1,241,256.601
Be III	ls.nf 1F* J=3	4–9	6	4	untested	4.0–9.0	+0.0002	0.0001	3	1,241,256.601
Be III	ls.ng 1G J=4	5–9	5	3	untested	5.0–9.0	+0.0002	0.0001	3	1,241,256.601
Be III	ls.nh 1H* J=5	6–9	4	2	untested	6.0–9.0	+0.0000	0.0000	3	1,241,256.601
Be III *	ls.ni 1I J=6	7–8	2	1	untested	7.0–8.0	+0.0000	0.0000	3	1,241,256.601
Be III	ls.np 3P* J=1	2–10	9	7	untested	2.0–10.0	+0.0429	0.0001	3	1,241,256.601
Be III	ls.ns 1S J=0	2–10	9	7	untested	1.9–9.9	+0.0508	0.0003	3	1,241,256.601
Be III	ls.ns 3S J=1	2–10	9	7	untested	1.9–9.9	+0.1302	0.0028	3	1,241,256.601
Be IV	nd 2D J=3/2	3–10	8	6	untested	3.0–10.0	+0.0002	0.0000	4	1,756,018.810
Be IV	nf 2F* J=5/2	4–10	7	5	untested	4.0–10.0	+0.0001	0.0000	4	1,756,018.810
Be IV	ng 2G J=7/2	5–10	6	4	untested	5.0–10.0	+0.0001	0.0000	4	1,756,018.810
Be IV	nh 2H* J=9/2	6–10	5	3	untested	6.0–10.0	+0.0000	0.0000	4	1,756,018.810
Be IV	ni 2I J=11/2	7–10	4	2	untested	7.0–10.0	+0.0000	0.0000	4	1,756,018.810
Be IV	nk 2K* J=13/2	8–10	3	1	untested	8.0–10.0	+0.0000	0.0000	4	1,756,018.810
Be IV	np 2P* J=1/2	3–10	8	6	untested	3.0–10.0	+0.0004	0.0000	4	1,756,018.810
Be IV	ns 2S J=1/2	3–10	8	6	untested	3.0–10.0	+0.0004	0.0000	4	1,756,018.810
Mg III *	nd 2[1/2]* J=1	4–5	2	1	untested	3.9–4.9	+0.0934	0.0014	3	646,402.000
Mg III	nd 2[3/2]* J=1	4–9	6	4	untested	4.0–9.0	+0.0388	0.0017	3	646,402.000
Mg III *	nf 2[3/2] J=1	4–5	2	1	untested	4.0–5.0	+0.0111	0.0005	3	646,402.000
Mg III *	nf 2[3/2] J=2	4–5	2	1	untested	4.0–5.0	+0.0108	0.0004	3	646,402.000
Mg III *	nf 2[5/2] J=3	4–5	2	1	untested	4.0–5.0	+0.0039	0.0003	3	646,402.000
Mg III *	nf 2[7/2] J=3	4–5	2	1	untested	4.0–5.0	+0.0000	0.0004	3	646,402.000
Mg III *	nf 2[7/2] J=4	4–5	2	1	untested	4.0–5.0	-0.0000	0.0004	3	646,402.000
Mg III *	nf 2[9/2] J=4	4–5	2	1	untested	4.0–5.0	+0.0073	0.0004	3	646,402.000
Mg III *	nf 2[9/2] J=5	4–5	2	1	untested	4.0–5.0	+0.0073	0.0004	3	646,402.000
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Mg III	ns 2[3/2]* J=1	4–6	3	1	untested	3.1–5.2	+0.8487	0.0050	3	646,402.000
Mg III *	ns 2[3/2]* J=2	4–5	2	1	untested	3.1–4.1	+0.8634	0.0030	3	646,402.000
Al III	nd 2D J=5/2	3–8	6	4	untested	2.9–7.9	+0.0644	0.0071	3	229,445.710
Al III	nf 2F* J=5/2	4–9	6	4	untested	4.0–9.0	+0.0044	0.0005	3	229,445.710
Al III	ng 2G J=7/2	5–9	5	3	untested	5.0–9.0	+0.0010	0.0001	3	229,445.710
Al III	nh 2H* J=9/2	6–9	4	2	untested	6.0–9.0	+0.0003	0.0000	3	229,445.710

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Al III	np 2P* J=1/2	3-7	5	3	untested	2.4-6.4	+0.6056	0.0128	3	229,445.710
Al III	ns 2S J=1/2	3-8	6	4	untested	2.1-7.1	+0.9049	0.0098	3	229,445.710
Si III	nd 1D J=2	3-9	7	5	untested	3.1-9.0	+0.0507	0.1195	3	270,139.300
Si III	nd 3D J=3	3-9	7	5	untested	2.8-8.8	+0.1977	0.0294	3	270,139.300
Si III	nf 1F* J=3	4-9	6	4	untested	3.9-9.0	+0.0240	0.1413	3	270,139.300
Si III	nf 3F* J=2	4-9	6	4	untested	4.0-9.0	+0.0226	0.0272	3	270,139.300
Si III	nf 3F* J=3	4-9	6	4	untested	4.0-9.0	+0.0224	0.0275	3	270,139.300
Si III	nf 3F* J=4	4-9	6	4	untested	4.0-9.0	+0.0221	0.0280	3	270,139.300
Si III	ng 1G J=4	5-8	4	2	untested	5.0-8.0	+0.0266	0.0040	3	270,139.300
Si III	ng 3G J=3	5-9	5	3	untested	5.0-9.0	+0.0281	0.0047	3	270,139.300
Si III	ng 3G J=4	5-9	5	3	untested	5.0-9.0	+0.0281	0.0047	3	270,139.300
Si III	ng 3G J=5	5-9	5	3	untested	5.0-9.0	+0.0280	0.0047	3	270,139.300
Si III	nh 1H* J=5	6-9	4	2	untested	6.0-9.0	+0.0082	0.0007	3	270,139.300
Si III	nh 3H* J=5	7-9	3	1	untested	7.0-9.0	+0.0086	0.0004	3	270,139.300
Si III	nh 3H* J=6	6-9	4	2	untested	6.0-9.0	+0.0082	0.0007	3	270,139.300
Si III	ni 1I J=6	7-9	3	1	untested	7.0-9.0	+0.0033	0.0002	3	270,139.300
Si III	ni 3I J=7	7-9	3	1	untested	7.0-9.0	+0.0033	0.0002	3	270,139.300
Si III	np 1P* J=1	4-7	4	2	untested	3.2-6.3	+0.7576	0.0165	3	270,139.300
Si III	np 3P* J=0	4-7	4	2	untested	3.2-6.3	+0.7563	0.0149	3	270,139.300
Si III	ns 1S J=0	4-8	5	3	untested	3.0-7.0	+1.0185	0.0017	3	270,139.300
Si III	ns 3S J=1	4-8	5	3	untested	2.9-7.0	+1.0646	0.0175	3	270,139.300
Ar II	3s2.3p4.(3P).nd 4D J=7/2	3-5	3	1	untested	2.2-4.4	+0.6930	0.0749	2	222,848.300
Ar II	3s2.3p4.(3P).ns 4P J=1/2	4-6	3	1	untested	2.2-4.4	+1.6821	0.0615	2	222,848.300
Ar II	3s2.3p4.(3P).ns 4P J=3/2	4-6	3	1	untested	2.2-4.3	+1.7246	0.0283	2	222,848.300
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ar II	3s2.3p4.(3P).ns 4P J=5/2	4-6	3	1	untested	2.2-4.3	+1.7463	0.0203	2	222,848.300
B IV	1s.nd 1D J=2	3-10	8	6	untested	3.0-10.0	+0.0007	0.0002	4	2,091,995.451
B IV	1s.nd 3D J=1	3-10	8	6	untested	3.0-10.0	+0.0021	0.0003	4	2,091,995.451
B IV	1s.nf 1F* J=3	4-9	6	4	untested	4.0-9.0	+0.0004	0.0005	4	2,091,995.451
B IV	1s.ng 1G J=4	5-9	5	3	untested	5.0-9.0	+0.0002	0.0004	4	2,091,995.451
B IV	1s.nh 1H* J=5	6-9	4	2	untested	6.0-9.0	-0.0002	0.0002	4	2,091,995.451
B IV *	1s.ni 1I J=6	7-8	2	1	untested	7.0-8.0	-0.0003	0.0003	4	2,091,995.451
B IV	1s.np 1P* J=1	3-10	8	6	untested	3.0-10.0	-0.0096	0.0003	4	2,091,995.451
B IV	1s.np 3P* J=0	3-10	8	6	untested	3.0-10.0	+0.0350	0.0001	4	2,091,995.451
B IV	1s.ns 1S J=0	3-10	8	6	untested	3.0-10.0	+0.0392	0.0004	4	2,091,995.451
B IV	1s.ns 3S J=1	3-10	8	6	untested	2.9-9.9	+0.1011	0.0009	4	2,091,995.451
B V	nd 2D J=3/2	3-10	8	6	untested	3.0-10.0	+0.0005	0.0002	5	2,744,111.380
B V	nf 2F* J=5/2	4-10	7	5	untested	4.0-10.0	+0.0004	0.0002	5	2,744,111.380
B V	ng 2G J=7/2	5-9	5	3	untested	5.0-9.0	+0.0003	0.0002	5	2,744,111.380
B V	nh 2H* J=9/2	6-9	4	2	untested	6.0-9.0	+0.0003	0.0001	5	2,744,111.380
B V	ni 2I J=11/2	7-9	3	1	untested	7.0-9.0	+0.0004	0.0001	5	2,744,111.380
B V *	nk 2K* J=13/2	8-9	2	1	untested	8.0-9.0	+0.0004	0.0001	5	2,744,111.380
B V	np 2P* J=1/2	3-10	8	6	untested	3.0-10.0	+0.0008	0.0002	5	2,744,111.380
B V	ns 2S J=1/2	3-10	8	6	untested	3.0-10.0	+0.0008	0.0002	5	2,744,111.380
Si IV	nd 2D J=5/2	3-8	6	4	untested	2.9-7.9	+0.0802	0.0079	4	364,093.100

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Si IV	nf 2F* J=5/2	4-8	5	3	untested	4.0-8.0	+0.0054	0.0007	4	364,093.100
Si IV	ng 2G J=7/2	5-8	4	2	untested	5.0-8.0	+0.0011	0.0001	4	364,093.100
Si IV *	nh 2H* J=9/2	6-7	2	1	untested	6.0-7.0	+0.0003	0.0001	4	364,093.100
Si IV	np 2P* J=1/2	3-8	6	4	untested	2.4-7.5	+0.5262	0.0121	4	364,093.100
Si IV	ns 2S J=1/2	3-8	6	4	untested	2.2-7.2	+0.7845	0.0094	4	364,093.100
P IV	nd 1D J=2	4-6	3	1	untested	3.8-5.8	+0.1991	0.0295	4	415,551.800
P IV	nd 3D J=1	4-6	3	1	untested	3.8-5.8	+0.2312	0.0239	4	415,551.800
P IV	nf 3F* J=2	4-6	3	1	untested	4.0-5.9	+0.0597	0.0170	4	415,551.800
P IV	np 1P* J=1	4-6	3	1	untested	3.3-5.4	+0.6991	0.0588	4	415,551.800
P IV	np 3P* J=0	4-6	3	1	untested	3.3-5.3	+0.6699	0.0079	4	415,551.800
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
P IV	ns 1S J=0	4-6	3	1	untested	3.1-5.1	+0.9018	0.0111	4	415,551.800
P IV	ns 3S J=1	4-7	4	2	untested	3.1-6.0	+0.9498	0.0073	4	415,551.800
C V	1s.nd 1D J=2	3-7	5	3	untested	3.0-7.0	+0.0108	0.0077	5	3,162,788.800
C V	1s.nd 3D J=1	3-7	5	3	untested	3.0-7.0	+0.0122	0.0078	5	3,162,788.800
C V	1s.nf 1F* J=3	4-6	3	1	untested	4.0-6.0	+0.0092	0.0042	5	3,162,788.800
C V	1s.nf 3F* J=3	4-6	3	1	untested	4.0-6.0	+0.0092	0.0041	5	3,162,788.800
C V	1s.ng 1G J=4	5-7	3	1	untested	5.0-7.0	+0.0152	0.0059	5	3,162,788.800
C V	1s.ng 3G J=4	5-7	3	1	untested	5.0-7.0	+0.0153	0.0059	5	3,162,788.800
C V *	1s.nh 1H* J=5	6-7	2	1	untested	6.0-7.0	+0.0186	0.0042	5	3,162,788.800
C V	1s.np 1P* J=1	2-7	6	4	untested	2.0-7.0	+0.0002	0.0084	5	3,162,788.800
C V	1s.np 3P* J=0	2-7	6	4	untested	2.0-6.9	+0.0384	0.0075	5	3,162,788.800
C V	1s.ns 1S J=0	2-7	6	4	untested	2.0-6.9	+0.0399	0.0079	5	3,162,788.800
C V	1s.ns 3S J=1	2-7	6	4	untested	1.9-6.9	+0.0927	0.0063	5	3,162,788.800
S V	3s.nd 1D J=2	4-6	3	1	untested	3.8-5.8	+0.1819	0.0275	5	586,610.400
S V	3s.nd 3D J=1	4-7	4	2	untested	3.8-6.8	+0.2170	0.0211	5	586,610.400
S V	3s.nf 1F* J=3	4-6	3	1	untested	4.0-6.0	+0.0002	0.0246	5	586,610.400
S V	3s.nf 3F* J=2	4-6	3	1	untested	4.0-5.9	+0.0637	0.0116	5	586,610.400
S V	3s.np 1P* J=1	4-7	4	2	untested	3.4-6.4	+0.5900	0.0229	5	586,610.400
S V	3s.np 3P* J=0	4-6	3	1	untested	3.4-5.3	+0.6221	0.0348	5	586,610.400
S V	3s.ns 1S J=0	4-7	4	2	untested	3.2-6.2	+0.8043	0.0144	5	586,610.400
S V	3s.ns 3S J=1	4-7	4	2	untested	3.2-6.1	+0.8428	0.0080	5	586,610.400
Fe XV *	nd 3D J=1	4-5	2	1	untested	3.9-4.9	+0.1278	0.0017	15	3,679,500.000
Fe XV	nf 1F* J=3	4-6	3	1	untested	4.0-6.5	-0.2367	0.2030	15	3,679,500.000
Fe XV *	nf 3F* J=2	4-5	2	1	untested	4.0-5.0	+0.0371	0.0016	15	3,679,500.000
Fe XV *	np 1P* J=1	4-5	2	1	untested	3.7-4.7	+0.2872	0.0017	15	3,679,500.000
Ca IX	nd 1D J=2	4-6	3	1	untested	3.9-5.8	+0.1572	0.0090	9	1,520,640.000
Ca IX *	nf 1F* J=3	4-5	2	1	untested	4.0-5.0	+0.0124	0.0050	9	1,520,640.000
Ca IX	np 1P* J=1	4-6	3	1	untested	3.6-5.6	+0.4121	0.0041	9	1,520,640.000
Ca IX *	ns 3S J=1	4-5	2	1	untested	3.4-4.4	+0.5725	0.0078	9	1,520,640.000
Ti XI *	nf 1F* J=3	4-5	2	1	untested	4.0-5.0	+0.0129	0.0041	11	2,137,900.000
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ti XI	np 1P* J=1	4-7	4	2	untested	3.6-6.7	+0.3273	0.0161	11	2,137,900.000
Ti XI *	ns 1S J=0	4-5	2	1	untested	3.5-4.5	+0.4738	0.0070	11	2,137,900.000
Ti XI *	ns 3S J=1	4-5	2	1	untested	3.5-4.5	+0.4983	0.0067	11	2,137,900.000

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ti III *	nd 1F J=3	4-5	2	1	untested	3.2-4.2	+0.7987	0.0056	3	223,827.100
Ti III *	nd 3D J=1	4-5	2	1	untested	3.2-4.2	+0.7899	0.0076	3	223,827.100
Ti III *	np 1D* J=2	4-5	2	1	untested	2.6-3.6	+1.4159	0.0063	3	223,827.100
Ti III *	np 3D* J=1	4-5	2	1	untested	2.6-3.6	+1.4039	0.0025	3	223,827.100
Ti III *	np 3F* J=2	4-5	2	1	untested	2.6-3.6	+1.3956	0.0072	3	223,827.100
Ti III *	ns 1D J=2	4-5	2	1	untested	2.3-3.3	+1.6726	0.0013	3	223,827.100
Ti III *	ns 3D J=1	4-5	2	1	untested	2.3-3.3	+1.6901	0.0041	3	223,827.100
Ca II	ns 2S J=1/2	4-6	3	1	untested	2.1-4.2	+1.8338	0.0183	2	95,751.880
Ca II	nd 2D J=3/2	3-16	14	12	untested	2.3-15.4	+0.6341	0.0154	2	95,751.870
Ca II	nf 2F* J=5/2	4-10	7	5	untested	4.0-10.0	+0.0248	0.0029	2	95,751.870
Ca II	ng 2G J=7/2	5-9	5	3	untested	5.0-9.0	+0.0048	0.0004	2	95,751.870
Ca II *	nh 2H* J=9/2	8-10 [†]	2	1	untested	8.0-10.0	+0.0016	0.0000	2	95,751.870
Ca II	np 2P* J=1/2	4-6	3	1	untested	2.5-4.5	+1.4775	0.0207	2	95,751.870
Ca II	ns 2S J=1/2	4-10	7	5	untested	2.1-8.2	+1.8192	0.0175	2	95,751.870
Cd II	nd 2D J=3/2	5-13	9	7	untested	3.1-11.1	+1.8942	0.0159	2	136,374.740
Cd II	nf 2F* J=5/2	4-9	6	4	untested	4.0-8.9	+0.0489	0.0092	2	136,374.740
Cd II	ng 2G J=7/2	5-11	7	5	untested	5.0-11.0	+0.0076	0.0004	2	136,374.740
Cd II	np 2P* J=1/2	5-12 [†]	7	5	untested	2.2-9.3	+2.7444	0.0319	2	136,374.740
Cd II	ns 2S J=1/2	5-13	9	7	untested	1.8-9.9	+3.1175	0.0330	2	136,374.740
Ca I	nd 1D J=2	4-9	6	4	untested	3.0-8.6	+0.9084	0.2360	1	49,305.950
Ca I	nd 3D J=1	4-9	6	4	untested	3.1-8.4	+0.8917	0.1405	1	49,305.950
Ca I	nf 1F* J=3	4-8	5	3	untested	4.0-7.9	+0.0654	0.0190	1	49,305.950
Ca I	nf 3F* J=2	4-9	6	4	untested	3.9-8.9	+0.0893	0.0059	1	49,305.950
Ca I	np 1P* J=1	5-10	6	4	untested	3.0-8.2	+1.8902	0.1793	1	49,305.950
Ca I	ns 1S J=0	5-11	7	5	untested	2.6-8.7	+2.3548	0.0350	1	49,305.950
Ca I	ns 3S J=1	5-11	7	5	untested	2.5-8.6	+2.4639	0.0227	1	49,305.950
Ar I	nd 2[7/2]* J=4	3-6	4	2	untested	2.8-5.6	+0.3516	0.0715	1	127,198.600

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ar I	np 2[1/2] J=1	4-7	4	2	untested	2.2-5.2	+1.7855	0.0204	1	127,198.600
Ar I	np 2[3/2] J=1	4-7	4	2	untested	2.3-5.3	+1.7107	0.0123	1	127,198.600
Ar I	np 2[5/2] J=3	4-7	4	2	untested	2.2-5.2	+1.7437	0.0119	1	127,198.600
Ar I	ns 2[3/2]* J=1	4-7	4	2	untested	1.8-4.8	+2.1701	0.0138	1	127,198.600
Ar I	ns 2[3/2]* J=2	4-7	4	2	untested	1.8-4.8	+2.1886	0.0132	1	127,198.600
Cd I	nd 1D J=2	5-15 [†]	10	8	untested	2.9-12.8	+2.1886	0.0288	1	72,540.050
Cd I	nd 3D J=1	5-11	7	5	untested	2.9-8.9	+2.0898	0.0049	1	72,540.050
Cd I	nf 3F* J=3	4-10	7	5	untested	4.0-10.0	+0.0346	0.0034	1	72,540.050
Cd I	np 1P* J=1	6-12	7	5	untested	2.9-8.9	+3.0515	0.0005	1	72,540.050
Cd I	np 3P* J=0	6-10	5	3	untested	2.8-6.8	+3.1814	0.0184	1	72,540.050
Cd I	ns 1S J=0	6-15	10	8	untested	2.4-11.4	+3.5926	0.0131	1	72,540.050
Cd I	ns 3S J=1	6-16	11	9	untested	2.3-12.3	+3.6677	0.0188	1	72,540.050
Ga II	nd 1D J=2	4-7	4	2	untested	3.3-6.1	+0.8475	0.1127	2	165,465.800
Ga II	nd 3D J=1	4-7	4	2	untested	2.9-5.9	+1.0630	0.0133	2	165,465.800
Ga II	nf 1F* J=3	4-7	4	2	untested	4.0-6.9	+0.0598	0.0070	2	165,465.800
Ga II	nf 3F* J=2	4-7	4	2	untested	3.9-6.9	+0.0617	0.0080	2	165,465.800
Ga II	ng (1/2,7/2) J=3	5-8	4	2	untested	5.0-8.0	+0.0146	0.0012	2	165,465.800
Ga II	ng (1/2,9/2) J=5	5-8	4	2	untested	5.0-8.0	+0.0142	0.0010	2	165,465.800
Ga II	ns 1S J=0	5-8	4	2	untested	2.7-5.8	+2.2610	0.0138	2	165,465.800

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ga II	ns 3S J=1	5–8	4	2	untested	2.6–5.7	+2.3228	0.0169	2	165,465.800
Hg II	nd 2D J=3/2	6–11	6	4	untested	3.1–8.1	+2.8758	0.0227	2	151,269.200
Hg II	nf 2F* J=7/2	5–8	4	2	untested	4.0–6.9	+1.0622	0.0083	2	151,269.200
Hg II	ng 2G J=9/2	5–9	5	3	untested	5.0–9.0	+0.0053	0.0028	2	151,269.200
Hg II	np 2P* J=1/2	6–8	3	1	untested	2.1–4.2	+3.8196	0.0624	2	151,269.200
Hg II	ns 2S J=1/2	6–11	6	4	untested	1.7–6.8	+4.1850	0.0515	2	151,269.200
P III	nd 2D J=5/2	3–6	4	2	untested	2.8–5.7	+0.2754	0.0451	3	243,573.700
P III	ng 2G J=7/2	5–8	4	2	untested	5.0–8.0	+0.0197	0.0011	3	243,573.700
P III	nh 2H* J=9/2	6–8	3	1	untested	6.0–8.0	+0.0024	0.0012	3	243,573.700
P III	np 2P* J=1/2	4–6	3	1	untested	3.1–5.1	+0.8886	0.0128	3	243,573.700
P III	ns 2S J=1/2	4–9	6	4	untested	2.8–7.8	+1.1692	0.0144	3	243,573.700
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ba II	nd 2D J=3/2	5–25 [†]	14	12	untested	2.4–22.6	+2.4147	0.0516	2	80,687.900
Ba II	nf 2F* J=5/2	4–12	9	7	untested	3.7–11.1	+0.7559	0.1645	2	80,687.900
Ba II	ng 2G J=7/2	5–10	6	4	untested	5.0–10.0	+0.0185	0.0022	2	80,687.900
Ba II	np 2P* J=1/2	6–12	7	5	untested	2.7–8.8	+3.2408	0.0287	2	80,687.900
Ba II	ns 2S J=1/2	6–30 [†]	18	16	untested	2.3–26.4	+3.5984	0.0191	2	80,687.900
Fe XVI	nd 2D J=3/2	3–8	6	4	untested	2.9–7.9	+0.0750	0.0031	16	3,946,570.000
Fe XVI	nf 2F* J=5/2	4–8	5	3	untested	4.0–8.0	+0.0116	0.0036	16	3,946,570.000
Fe XVI	np 2P* J=1/2	3–6	4	2	untested	2.8–5.8	+0.2233	0.0059	16	3,946,570.000
Fe XVI	ns 2S J=1/2	3–7	5	3	untested	2.7–6.7	+0.3174	0.0152	16	3,946,570.000
N III	nd 2D J=3/2	3–12	10	8	untested	2.9–11.9	+0.0737	0.0124	3	382,672.000
N III	nf 2F* J=5/2	4–7	4	2	untested	4.0–7.0	+0.0281	0.0048	3	382,672.000
N III	ng 2G J=7/2	5–7	3	1	untested	5.0–7.0	+0.0022	0.0009	3	382,672.000
N III *	nh 2H* J=9/2	6–7	2	1	untested	6.0–7.0	-0.0026	0.0009	3	382,672.000
N III	np 2P* J=1/2	3–5	3	1	untested	2.7–4.7	+0.2917	0.0185	3	382,672.000
N III	ns 2S J=1/2	3–14	12	10	untested	2.5–13.3	+0.5660	0.0740	3	382,672.000
N I	nd 2F J=5/2	3–5	3	1	untested	3.0–5.0	+0.0273	0.0024	1	117,244.100
N I *	nd 4D J=1/2	3–4	2	1	untested	3.0–4.0	+0.0041	0.0041	1	117,244.100
N I	nd 4F J=3/2	3–5	3	1	untested	3.0–4.9	+0.0514	0.0035	1	117,244.100
N I	np 2D* J=3/2	3–5	3	1	untested	2.3–4.5	+0.6233	0.0955	1	117,244.100
N I	np 4D* J=1/2	3–5	3	1	untested	2.2–4.2	+0.7714	0.0136	1	117,244.100
N I	np 4P* J=1/2	3–5	3	1	untested	2.2–4.3	+0.7329	0.0169	1	117,244.100
N I	ns 2P J=1/2	3–6	4	2	untested	1.9–4.9	+1.1040	0.0116	1	117,244.100
N I	ns 4P J=1/2	3–6	4	2	untested	1.8–4.8	+1.1697	0.0190	1	117,244.100
S IV	nd 2D J=3/2	3–5	3	1	untested	2.8–4.7	+0.2649	0.0197	4	382,135.000
S IV	np 2P* J=1/2	4–7	4	2	untested	3.2–6.2	+0.7607	0.0167	4	382,135.000
S IV	ns 2S J=1/2	4–7	4	2	untested	3.0–6.0	+1.0264	0.0100	4	382,135.000
S VI	nd 2D J=3/2	3–9 [†]	6	4	untested	2.9–8.9	+0.0925	0.0069	6	710,194.700
S VI	nf 2F* J=5/2	4–10	7	5	untested	4.0–10.0	+0.0074	0.0011	6	710,194.700
S VI *	ng 2G J=7/2	5–6	2	1	untested	5.0–6.0	+0.0010	0.0001	6	710,194.700
S VI *	nh 2H* J=9/2	6–7	2	1	untested	6.0–7.0	+0.0003	0.0001	6	710,194.700
species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
S VI	np 2P* J=1/2	3–7	5	3	untested	2.6–6.6	+0.4228	0.0110	6	710,194.700
S VI	ns 2S J=1/2	3–8	6	4	untested	2.4–7.4	+0.6242	0.0083	6	710,194.700
Al IV *	nf 2[5/2] J=3	4–5	2	1	untested	4.0–5.0	+0.0349	0.0100	4	968,924.400
Al IV	ns 2[3/2]* J=2	3–5	3	1	untested	2.2–4.2	+0.7638	0.0042	4	968,924.400

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
B II	nd 1D J=2	3-7	5	3	untested	3.0-7.1	-0.0323	0.0111	2	202,691.900
B II	nd 3D J=1	3-6	4	2	untested	2.9-5.9	+0.0772	0.0138	2	202,691.900
B II	nf 3F* J=2	4-7	4	2	untested	4.0-6.9	+0.0397	0.0132	2	202,691.900
B II	ng 3G J=3	5-7	3	1	untested	5.0-7.1	-0.0401	0.0192	2	202,691.900
B II	np 1P* J=1	3-5	3	1	untested	2.7-4.6	+0.3284	0.0668	2	202,691.900
B II	np 3P* J=0	3-6	4	2	untested	2.7-5.8	+0.2231	0.0345	2	202,691.900
B II	ns 1S J=0	3-6	4	2	untested	2.6-5.6	+0.4303	0.0172	2	202,691.900
B II	ns 3S J=1	3-7	5	3	untested	2.5-6.5	+0.5211	0.0148	2	202,691.900
C III	nd 1D J=2	3-7	5	3	untested	3.0-7.0	+0.0067	0.0036	3	386,241.000
C III	nd 3D J=1	3-9	7	5	untested	2.9-8.9	+0.0816	0.0069	3	386,241.000
C III	nf 3F* J=2	4-7	4	2	untested	3.9-7.0	+0.0127	0.0398	3	386,241.000
C III	ng 3G J=3	5-7	3	1	untested	5.0-7.0	+0.0126	0.0021	3	386,241.000
C III	np 1P* J=1	3-8	6	4	untested	2.8-7.8	+0.1806	0.0527	3	386,241.000
C III	np 3P* J=0	3-6	4	2	untested	2.8-5.8	+0.1856	0.0217	3	386,241.000
C III	ns 1S J=0	3-5	3	1	untested	2.7-4.5	+0.3819	0.0499	3	386,241.000
C III	ns 3S J=1	3-7	5	3	untested	2.6-6.6	+0.3993	0.0137	3	386,241.000
O IV	nd 2D J=3/2	3-6	4	2	untested	2.9-5.9	+0.0681	0.0025	4	624,202.200
O IV	nf 2F* J=5/2	4-7	4	2	untested	4.0-6.9	+0.0408	0.0115	4	624,202.200
O IV *	ng 2G J=7/2	5-6	2	1	untested	5.0-6.0	-0.0028	0.0018	4	624,202.200
O IV *	nh 2H* J=9/2	6-7	2	1	untested	6.0-7.0	-0.0125	0.0030	4	624,202.200
O IV	np 2P* J=1/2	3-5	3	1	untested	2.7-4.8	+0.2462	0.0106	4	624,202.200
O IV	ns 2S J=1/2	3-5	3	1	untested	2.6-4.5	+0.4408	0.0072	4	624,202.200
F I	nd 4D J=7/2	3-6	4	2	untested	3.0-6.0	+0.0323	0.0008	1	140,518.700
F I	nd 4F J=9/2	3-6	4	2	untested	3.0-6.0	+0.0149	0.0012	1	140,518.700
F I *	np 4D* J=7/2	3-4	2	1	untested	2.2-3.2	+0.8308	0.0097	1	140,518.700
F I	ns 2P J=3/2	3-6	4	2	untested	1.8-4.9	+1.1857	0.0604	1	140,518.700

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
F I	ns 4P J=5/2	3-8	6	4	untested	1.7-6.7	+1.2800	0.0117	1	140,518.700
Ge III	nd 1D J=2	4-6	3	1	untested	3.2-5.1	+0.8820	0.0572	3	274,693.000
Ge III	nd 3D J=1	4-6	3	1	untested	3.0-5.0	+1.0025	0.0189	3	274,693.000
Ge III *	ng 3G J=3	5-6	2	1	untested	5.0-6.0	+0.0181	0.0010	3	274,693.000
Ge III	ns 1S J=0	5-7	3	1	untested	3.0-5.0	+2.0116	0.0057	3	274,693.000
Ge III	ns 3S J=1	5-8	4	2	untested	2.9-6.0	+2.0576	0.0162	3	274,693.000
K I	nd 2D J=5/2	3-13	11	9	untested	2.9-12.7	+0.2460	0.0380	1	35,010.600
K I	nf 2F* J=5/2	4-12	9	7	untested	4.0-12.0	+0.0112	0.0030	1	35,010.600
K I	np 2P* J=1/2	4-15	12	10	untested	2.2-13.3	+1.7268	0.0133	1	35,010.600
K I	ns 2S J=1/2	4-18	15	13	untested	1.8-15.8	+2.1912	0.0111	1	35,010.600
O III	nd 1D* J=2	3-5	3	1	untested	2.9-4.9	+0.1196	0.0040	3	443,325.200
O III *	nd 3D* J=1	3-4	2	1	untested	2.9-3.9	+0.0791	0.0043	3	443,325.200
O III	nd 3F* J=2	3-5	3	1	untested	2.9-4.9	+0.1290	0.0090	3	443,325.200
O III *	np 3D J=1	3-4	2	1	untested	2.6-3.6	+0.4221	0.0073	3	443,325.200
O III	ns 1P* J=1	3-5	3	1	untested	2.4-4.4	+0.5852	0.0048	3	443,325.200
O III	ns 3P* J=0	3-5	3	1	untested	2.4-4.4	+0.6250	0.0048	3	443,325.200
C I *	nd 1F* J=3	3-4	2	1	untested	3.0-4.0	+0.0320	0.0059	1	90,937.500
C I *	nd 3D* J=1	3-4	2	1	untested	2.9-3.9	+0.0638	0.0097	1	90,937.500
C I *	np 3D J=1	3-4	2	1	untested	2.3-3.3	+0.7201	0.0073	1	90,937.500
C I *	np 3P J=0	3-4	2	1	untested	2.4-3.4	+0.6284	0.0046	1	90,937.500

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
C I *	np 3S J=1	3–4	2	1	untested	2.3–3.3	+0.6641	0.0048	1	90,937.500
C I	ns 1P* J=1	3–5	3	1	untested	1.9–3.9	+1.0532	0.0037	1	90,937.500
C I	ns 3P* J=0	3–5	3	1	untested	1.9–3.9	+1.0925	0.0127	1	90,937.500
P II *	np 3P J=0	4–5	2	1	untested	2.8–3.8	+1.1670	0.0004	2	159,929.200
P II	ns 3P* J=0	4–6	3	1	untested	2.4–4.4	+1.5468	0.0075	2	159,929.200
S III *	ns 3P* J=0	4–5	2	1	untested	2.7–3.7	+1.2846	0.0049	3	281,130.000

596 channel rows across 28 elements · 2,269 interior cells parsed.

element	channels
Al	19
Ar	50
B	32
Ba	27
Be	30
Bi	22
C	36
Ca	18
Cd	12
F	5
Fe	8
Ga	13
Ge	5
He	16
Hg	5
K	8
Li	36
Mg	37
N	18
Na	12
Ne	44
O	12
P	14
S	18
Sc	11
Si	67
Ti	11
Zn	10

III • THE MECHANISMS

How a value is got where no capture reaches. Each is measured on pairs where both cells are known.

mechanism	the move	what it gives
the ground-state derivation	none	the multiplicity, and the bound B
the isoelectronic ladder	$(Z, c) \rightarrow (Z+1, c+1)$	δ along a sequence
the charge ladder	$c \rightarrow c+1$ at fixed Z	δ falls with charge
ℓ -collapse	$\ell \rightarrow \ell+1$	δ falls with ℓ

mechanism	the move	what it gives
the exchange split	singlet $\leftrightarrow$ triplet	the triplet defect exceeds the singlet at ns
the Janet collapse	a lookup in the periodic table	whether a $p = 0$ channel is large
Seaton's formula	none	$\delta_0 = 3\alpha/K(\ell)$ for a non-penetrating series

The isoelectronic ladder

Along a sequence of fixed electron count, a channel's defect follows $\delta(c) = A + B \cdot \ln(c+1)/c$. Fitted per sequence on the 52 with three or more charge states it reaches a median rms of **0.0083** — the tightest relation the compendium holds.

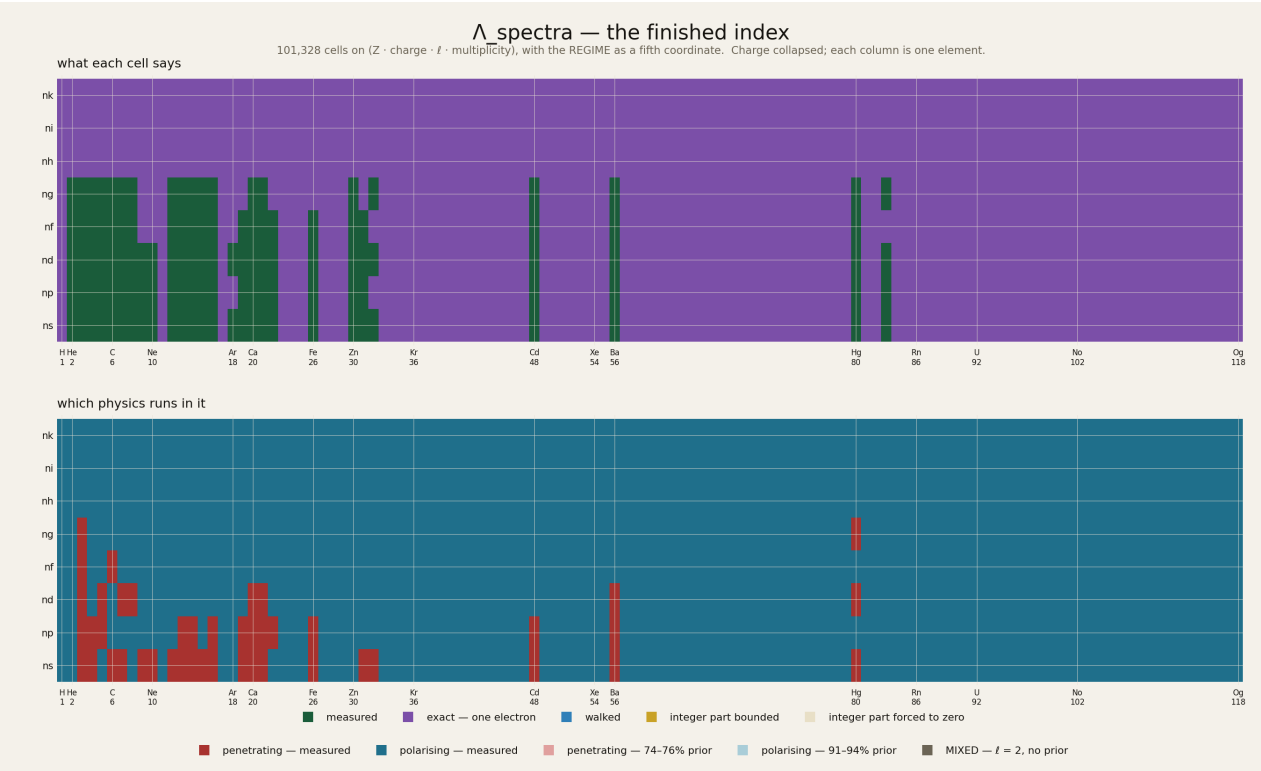
And **A is fixed rather than fitted**. As charge grows at constant electron count the core collapses to a point and the ion becomes hydrogenic, so $\delta \rightarrow 0$. That is the periodic table's own edge, and setting $A \equiv 0$ leaves one parameter.

A channel is a curve

$$\delta(n) = \delta_0 + \delta_2/(n - \delta_0)^2$$

δ_0 is the limit value the index carries; δ_2 is the curvature. Fitted on 274 channels, the second term removes **48%** of what a constant defect leaves as scatter.

And the sign of **δ_2 names the regime**: positive for core penetration, negative for core polarisation. By orbital: **s 76% positive, p 74%, d 37%, f 6%, g 7%**. The sign rule is NIST's own, stated in the reference the levels come from. What the compendium adds is the measurement across 274 channels and the finding that $|\delta_2|$ predicts a channel's scatter at $R^2 = 0.761$.



The index

IV • THE SOURCES

B.1 Sources compilation spectra drawn Kaufman & Martin Al I, Al II 1991, JPCRD 20, 775 Kramida & Martin Be I 1997, JPCRD 26, 1185 NIST ASD Ar II, Be II, Bi I, C II, Ca II, Cd II, Ga I, He I, He II, Hg II, K II, Li I, Li II, Mg II, N II, Na II, Ne I ($3/2$), Ne I ($1/2$), Si I, Si II, Zn II Sansonetti 2008, Na I JPCRD 37, 1659 Sansonetti 2008, KI JPCRD 37, 7 **Additional sources used for exotic systems and collective quantities:** Hori *et al.*, *Nature** **475**, 484 (2011) and Phys. Rev. Lett. **96**,

243401 (2006), both accessed via CODATA 2010 Table XII (arXiv:1203.5425) and arXiv:1304.4330 · Korobov, Phys. Rev. A* 77, 042506 (2008) · Singer, Stanojevic, Weidemüller & Côté, J. Phys. B* 38, S295 (2005) · Sugar & Corliss 1985, JPCRD* 14 Suppl. 2 · Sugar & Musgrove 1990, JPCRD* 19, 527 and 1995, JPCRD* 24, 1803. B.2 Channels **The 133 channel rows are lifted to the Spectra Compendium.** Each carries its species, channel, n-range, level count, interior cells, bracket, v range, mean quantum defect, its spread, Z_eff and the series limit. *Reproduced in the book only where one carries an argument; the rest are data and live where data belongs.* **THE METHOD 1.6 — SPECTRA COMPENDIUM**, generated by `spectra.py` from this source, so it cannot drift. 133 rows, 23 elements, 869 interior cells, bracket 869 of 869. **Totals: 153 channels, 1105 cells, bracket 1105/1105. And the table above holds 133 rows summing to 869 interior cells, bracket 869/869.** Twenty channels and 236 cells are **stated and not tabulated**. The bracket is internally consistent either way, so this is not a broken calculation — it is a total counting channels the table **does not list**. The additional sources named above for exotic systems and collective quantities carry sources and no rows, and twenty such channels would close the gap exactly. *Disclosed rather than reconciled: reconciling it needs the twenty rows, which are owed.* Registers 630–631. B.3 Flagged channels Channels whose δ spread exceeds 0.25 or whose V departs from $4v/3$ by more than 5% are flagged. Every flag in this collection has an identified cause; none is unexplained. specie channel δ cause s spread Al I $3s^2nd\ ^2D\ 0.630$ compilers relabelled this series; y^2D removed as a perturber Al II $3snd\ ^1D\ 0.390$ local perturber Al II $3snf\ ^3F^o$

Published values the coordinates are checked against

source	what it gives
NIST Atomic Spectra Database	the levels every capture is fitted to
Theodosiou, Inokuti & Manson, At. Data Nucl. Data Tables 35, 473 (1986)	asymptotic quantum defects of s, p, d and f orbitals for all ionisation stages of all ions with $Z \leq 50$, Hartree–Slater
Manson, Inokuti & Theodosiou, OSTI 7104964	the isoelectronic, isonuclear and isoionic pictures
Seaton 1958; Drake & Swainson 1991	the polarisation formula
Peper <i>et al.</i> 2019	K I to eight digits
Freeman & Kleppner 1976	Na I ng
arXiv:1706.06237	Cs I np at n = 70–100
arXiv:2508.06733	actinide defects, Z = 89–103
arXiv:2502.20961	Cs ⁺ dipole and quadrupole polarisabilities
ARC Alkali Rydberg Calculator	an independent check on K I

The full bibliography, with what each source was used for, is in LITERATURE.mcd.

The four capture groups

Register 1296 marked these as owed. Counts below are read from COORDINATES.tsv at build time, not transcribed.

group	species	cells	measured	exact	computed
Kr-core	Kr	512	0	8	504
Cd/In	Cd/In	1,384	13	16	1,355
La/Ce	La/Ce	1,632	0	16	1,616
Hg-core	Hg	1,168	5	8	1,155

4,696 cells across six species, every one spanning l = 0-7 and the full charge ladder from 1 to Z.

What the counts say, measured on this compendium's own basis

Section 0 states that accuracy is quoted against the VERIFIED-PLUS-POSSIBLE set and not against the whole index. The same basis is used here.

The four groups hold 18 reachable cells of the index's 358 - 5.0% of the reachable set on 4.5% of the cells. They are therefore slightly BETTER than representative, not worse.

On the raw grade column they carry 18 `measured` cells of 4,696 - Cd/In twelve and Hg five, with **Kr-core and La/Ce carrying none**. That ratio is not a fact about these groups: the whole index is 98.8% computed, which section 0's table already states. The filler is the index's condition, not these captures' shortcoming.

Why each was captured

- **Kr-core** - the $n = 4$ closure; its 5s manifold is still one level short - **Cd/In** - the d10 closure and the first post-d p-electron - **La/Ce** - THE JANET ANOMALY ITSELF - La opens 5d, Ce is where 4f appears - **Hg-core** - the 5d closure at high Z, where the relativistic wall stands

La/Ce is the group that earns its place. The other three fill coordinate space; **La/Ce is where the filling order itself breaks**, and holding it in the index lets the anomaly be stated as a coordinate fact rather than a footnote.